



BENCHMARKING SYNTHETIC TABULAR DATA GENERATORS FOR STRUCTURAL COHERENCE OF BEHAVIORAL AND EDUCATIONAL DATASETS

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Code available at <https://github.com/mithunthakkar8/Summer-Scholarship-Project>

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1. Abstract

Synthetic tabular data is increasingly used to share private data in behavioural and educational settings. However, its suitability for structural equation modelling (SEM) remains unclear. This study asks if contemporary synthetic data generators preserve the measurement, structural, and inferential properties required for valid SEM-based analysis.

A key obstacle is that most synthetic data generators were built for machine learning use cases with an explicit target variable. SEM-structured behavioural data instead contains only interrelated latent constructs. Naive substitutes such as PCA scores failed to preserve structural relationships. We resolved this by using the SEM-derived composite score of the endogenous construct as target. This is theoretically grounded since endogenous constructs act like dependent variables in linear SEM.

Using PISA 2022 and TIMSS 2023 (Singapore) - we benchmark synthetic tabular data generators. We use a unified SEM-oriented evaluation pipeline, which assesses distributional fidelity, downstream utility (SEM metrics) and privacy risk on multiple synthetic data replications.

To our knowledge, no study has systematically benchmarked generative architectures against SEM structural coherence criteria on behavioural data.

Results show that diffusion-based generators achieve strong preservation of structural relationships and global SEM fit, while LLM-based generators are sensitive to model size and tuning. In small-sample settings, certain diffusion models show substantial privacy leakage, whereas well-tuned LLMs perform better on measurement and structural criteria.

These findings highlight trade-offs between fidelity, stability, and privacy in synthetic data generation for SEM. The proposed evaluation framework provides practical guidance to deploy synthetic tabular data in behavioural and educational modelling.

2. Introduction

Behavioural, psychological, and educational research increasingly relies on large-scale individual-level datasets, such as international assessments, longitudinal school surveys, and questionnaire-based studies. These datasets include high-stakes decisions and are often sensitive since they contain detailed information about students, teachers, families, or patients. Access is hence tightly controlled, and there is growing concern that traditional de-identification techniques are insufficient to prevent re-identification or linkage attacks. (Liu et al., 2025; van Kesteren, 2024)

2.1 Synthetic Data for Behavioural and Educational Research

Synthetic data generation is increasingly being used as a promising strategy for data utility with privacy protection. Instead of sharing raw data, institutions can release artificially generated datasets that include the joint distribution of the original data while it reduces the risk of exposing real individuals.

2.2 New Class of Synthetic Data Generators

The rapid progress of large language models (LLMs) has motivated their use for tabular synthetic data generation. In parallel, diffusion-based generative models have emerged as a strong alternative paradigm for tabular data synthesis.

However, despite these advances, existing studies on both LLM-based and diffusion-based tabular data generation predominantly evaluate performance using distributional similarity metrics (e.g., marginal distributions, correlations) or downstream predictive utility (e.g., classification accuracy). To date, there is limited evidence that these models preserve latent variable structures, measurement models, or structural equation relationships that are central to theory-driven analyses in behavioural research.

2.3 Gap: Structural Coherence Has Not Been Systematically Benchmarked

Across both classical and synthetic tabular data literature, we need high-fidelity synthetic data that satisfy stricter utility and privacy criteria, especially for sensitive domains such as health and education (Liu et al., 2025; van Kesteren, 2024). Yet most evaluation frameworks still emphasise relatively shallow metrics: column-wise distributions, pairwise correlations, or model performance on simple classification/regression tasks.

For behavioural and human-centred datasets, these metrics are insufficient because they do not capture the structural layer of the data:

- **Factor structure:** stability of factor loadings and reliability for latent constructs.
- **Structural paths:** preservation of standardised regression coefficients between latent variables and covariates.
- **Latent correlations:** retention of theoretically important associations between constructs (e.g., self-concept and achievement).
- **Global SEM fit:** indices such as CFI, TLI, RMSEA, and SRMR that summarise how well the hypothesised model fits the covariance structure.

In this report, we refer to this bundle of properties as structural coherence. A structurally coherent synthetic dataset is one in which a pre-specified SEM fitted to synthetic data

recovers the key measurement and structural relationships observed in the real data, within acceptable tolerance.

There is currently no systematic benchmark that evaluates LLM-based and Diffusion Models-based synthetic data generators on structural coherence for behavioural or educational datasets. Existing LLM work focuses on generic tabular benchmarks, while SEM-oriented evaluations remain largely unexplored.

2.4 Research Aims and Contributions

This project addresses that gap by benchmarking LLM-based and Diffusion-based synthetic data generators on behavioural and human-centred datasets with focus on distributional, SEM and privacy focus. Concretely, the research:

1. **Develops a practical method for aligning SEM latent constructs with training requirements of diverse generative models**
2. **Generates synthetic data using LLM-based and Diffusion-based synthetic tabular generators on behavioural and educational data**
3. **Develops a structural coherence evaluation framework** that includes distribution, factor loadings, discriminant validity, reliability (α , CR, AVE), latent R^2 , latent correlations, structural path coefficients, privacy and global SEM fit indices.
4. **Applies this framework on Real data and Synthetic data**
5. **Quantifies structural coherence**
6. **Identifies common failure modes**
7. **Provides practical recommendations** for researchers seeking to use synthetic data in behavioural and educational contexts, including when LLM-based generators are appropriate, when diffusion-based generators are appropriate and when classical methods suffice, and what additional safeguards or post-processing steps may be needed.

2.5 Report Structure

This report is organised into five main parts. Following the introduction, the background and related work section reviews prior research on synthetic tabular data generation in behavioural and educational contexts, covering classical approaches, LLM-based and diffusion-based methods, structural coherence in behavioural science, and existing gaps in evaluation practices. The methodology section then provides a unified synthetic data distribution and SEM evaluation pipeline which includes dataset descriptions, preprocessing procedures, synthetic data generation methods, structural coherence evaluation, and reproducibility considerations. The experiments and results section

reports findings across variety of relevant metrics. The discussion section then discussing about interpretation of these results and provides recommendations.

3. Background and Related Work

Most existing synthetic data evaluations focus on distributional similarity, correlation fidelity, or downstream utility with focus on machine learning prediction (Borisov et al., 2022; Xu et al., 2019). These provide a necessary baseline but are insufficient for behavioural and educational research. That is because conclusions in such datasets depend on the stability of latent constructs and their interrelationships. This limitation necessitates the need for structure-aware evaluation frameworks that explicitly assess measurement quality, structural relationships, and global model coherence.

3.2 LLM-Based and Transformer-Based Tabular Generators

Recent advances in large language models (LLMs) have directed a shift in synthetic tabular data generation toward transformer-based approaches. LLM generators treat tabular records as sequences of tokens which helps leverage the contextual modelling capabilities of pretrained language models for synthesis of new data points (Borisov et al., 2022). This paradigm helps with flexible handling of heterogeneous feature types and complex dependencies. This has resulted in growing interest in LLMs as general-purpose tabular data generators.

A foundational method in this domain is GReaT (Generation of Realistic Tabular Data), which portrayed that autoregressive language models can be fine-tuned on tabular data and can generate realistic synthetic rows. This is done by sequentially sampling attributes conditioned on features that were generated previously (Borisov et al., 2022). GReaT established the feasibility of reframing tabular synthesis as a language-modelling task. It showed competitive performance relative to classical baselines such as CTGAN across standard similarity and downstream utility metrics.

Subsequent work has proposed refinements to the LLM-based framework. PredLLM (Nguyen et al., 2024) retains feature–target relationships in synthetic data via a prediction oriented objective. This improved similarity metrics and predictive performance. TabuLa trains a randomly initialised model LLM and restructures tokenisation to reduce sequence length of generated data (Zhao et al., 2025). REaLTabFormer was one of the first tabular data synthesis transformer architectures (Solatorio & Dupriez, 2023). TapTap furthered this through large-scale pretraining on tabular corpora which provided transformer models with general tabular knowledge prior to task-specific fine-tuning (T. Zhang et al., 2023).

3.3 Diffusion-Based Tabular Generators

Diffusion-based generative models have parallelly emerged as an alternative solution for synthetic tabular data generation. They offer a probabilistic framework which is different from both autoregressive LLM-based methods and GAN-based approaches. Diffusion models generate data by iteratively denoising samples that are drawn from a simple noise distribution through a learned reverse diffusion process (Ho et al., 2020). Examples of tabular data generators that adapt this framework as discussed below.

TabDDPM was one of the first adaptations of diffusion models to tabular data generation. This reformulated tabular generation as a denoising diffusion probabilistic process in a continuous latent space (Kotelnikov et al., 2023). It encodes mixed-type variables into continuous representations, while stable learning of high-dimensional joint distributions is enabled. TabDiff introduces feature-wise learnable diffusion processes within a joint continuous-time diffusion framework. This helps numerical or categorical variables to be modelled according to their own distributional characteristics (Shi et al., 2024). TabSyn is a VAE and diffusion hybrid model. It embeds tabular data into a structured latent space prior to generation. (H. Zhang et al., 2023).

3.4 Key Features of Data Generators

Below is a table illustrating the key features offered by all data generators we have discussed above:

Table 1
Comparison of Synthetic Tabular Data Generation Techniques and Their Core Modelling Principles

Technique	Model family	Core idea	Paper
CTGAN	GAN (Conditional Tabular GAN)	Generator + discriminator trained adversarially with conditional sampling	(Xu et al., 2019)
		Row serialization with autoregressive language modelling; optional start-column distribution bias and numeric token splitting	(Borisov et al., 2022)
GReaT	LLM (autoregressive transformer)	Row serialization with autoregressive language modelling; optional start-column distribution bias and numeric token splitting	(T. Zhang et al., 2023)
TapTap	LLM (autoregressive transformer)	Row serialization with autoregressive language modelling; optional start-column distribution bias and numeric token splitting	
Tabula	LLM fine-tuning	Lightweight GPT fine-tuning on tabular text	(Zhao et al., 2025)
TabDiff	Diffusion (DDPM)	Iterative denoising of noise vectors to generate full rows	(Shi et al., 2024)

		jointly in continuous latent space	
TabSyn	Diffusion + VAE hybrid	Encode rows into latent embeddings, then apply diffusion for stable, sample-efficient synthesis	(H. Zhang et al., 2023)
PredLLM	LLM (predictive formulation)	Autoregressive generation with target-oriented token ordering to improve supervised predictive utility	(Nguyen et al., 2024)
REaLTabFormer	Transformer (BERT/GPT hybrid)	Bidirectional + autoregressive transformer tailored for relational/tabular structure modelling	(Solatorio & Dupriez, 2023)

Note. LLM = large language model; DDPM = denoising diffusion probabilistic model; VAE = variational autoencoder.

3.5 Gaps in Existing Work

The existing literature demonstrates good progress in synthetic tabular data generation, including classical probabilistic models, LLM-based generators, and diffusion-based approaches. These methods have been evaluated on generic tabular benchmarks using metrics such as marginal distribution similarity, pairwise correlations, downstream predictive performance, and privacy risk.

A related but distinct strand of work applies SEM-oriented thinking to generative modelling. Structural Equation-VAE (Zhang et al., 2025) was introduced as a variational autoencoder with sub-encoders that are construct-aligned. This helps produce a latent space that mirrors the structure of a measurement model in SEM. This work evaluates disentanglement using metrics such as MIG, DCI, and SAP, on simulated data with known ground-truth latent structure. It does not evaluate synthetic data generation for downstream SEM analysis, and does not test on real behavioural or educational assessment data. To date, no study has systematically benchmarked trained generative architectures against SEM structural coherence criteria using behavioural or educational data.

This gap motivates the present study which involves benchmarking multiple classes of synthetic tabular generators. This involves using a structural coherence-oriented evaluation framework. This work aims to provide a systematic, theory-relevant assessment of synthetic data quality for human-centred and behavioural research.

3.6 Commercial Synthetic Data Platforms

Beyond academic and open-source, there exists a growing ecosystem of commercial platforms that offer end-to-end synthetic data solutions that target enterprise,

healthcare, and regulated data environments. These platforms typically prioritise usability, compliance, and rapid deployment and abstract away model-level details in favour of automated pipelines and dashboard-driven evaluation.

Tonic.ai is a widely adopted commercial platform that provides synthetic tabular data generation with a strong emphasis on privacy protection, regulatory compliance, and downstream application testing (Tonic.ai, 2023). Similarly, MOSTLY AI focuses on enterprise-grade synthetic data generation. It uses probabilistic and deep generative models and offer automated controls for utility-privacy trade-off. (Mostly AI, 2023).

Gretel.ai helps generate synthetic tabular, time-series, and text data. The platform supports developer-friendly workflows and configurable privacy settings. It provides APIs and evaluation tools designed for production workflows (Gretel.ai,2023).

Hazy's SynthData Studio offers synthetic data generation combined with governance, auditability, and compliance reporting features. These services are used by finance and public companies in Europe (Hazy, 2022).

These platforms demonstrate the practical maturity of synthetic data technologies, and their evaluation frameworks predominantly emphasise distributional similarity, privacy risk, and downstream task performance. They too do not focus on systematic assessment of measurement validity, latent construct preservation, and structural coherence - central to behavioural and educational research - remains largely outside the scope of current commercial offerings. This further motivates the need for academically grounded, SEM-oriented benchmarking frameworks such as the one adopted in this study.

4. Methodology

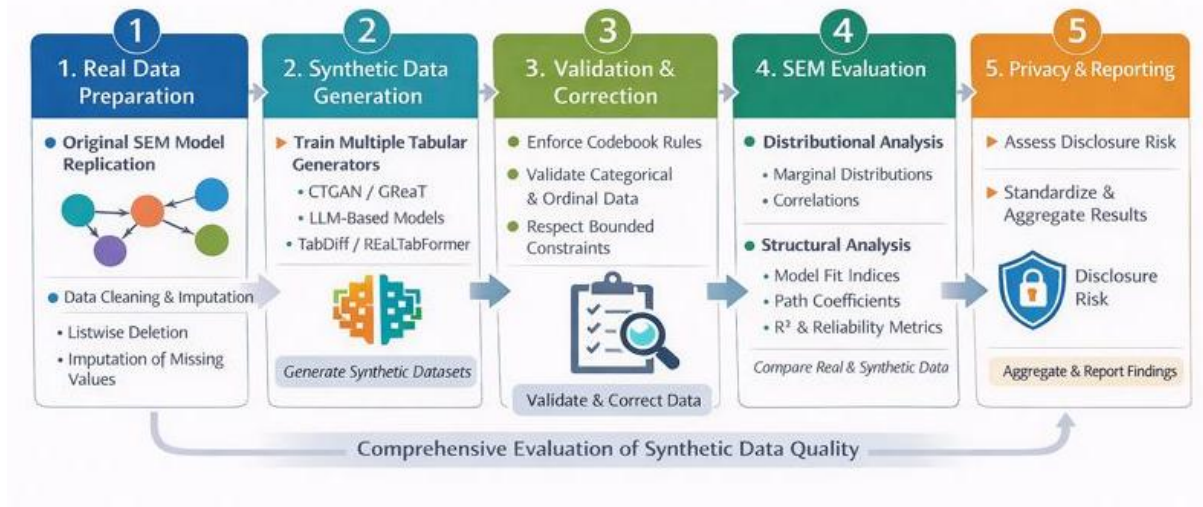
4.1 Overview of the Synthetic Data and SEM Evaluation Pipeline

Figure

1

Unified Synthetic Data Evaluation Pipeline for PISA 2022 and TIMSS 2023.

Synthetic Data Evaluation Pipeline for PISA 2022 and TIMSS 2023



This study implements a unified synthetic data evaluation pipeline grounded in structural equation modelling (SEM) to assess the suitability of synthetic tabular data for behavioural and educational research (see Figure 1). The pipeline is applied consistently across two large-scale international assessment contexts: PISA 2022 (Singapore) and TIMSS 2023 (Singapore), each using an established, theory-driven SEM specified in prior empirical research.

For each dataset, the original SEM is first replicated using the real data. This establishes a reference model against which all synthetic data evaluations are conducted. Prior to synthetic data generation, incomplete cases for SEM indicator blocks were removed using listwise deletion to ensure valid estimation of measurement and structural models. Missing values in non-indicator covariates were then imputed using simple, distribution-preserving strategies (median for continuous variables and mode for categorical variables). This ensured complete, generator-compatible datasets whilst maintaining consistency across real and synthetic evaluations. Synthetic datasets are then generated at the same sample size as the original data using multiple classes of tabular generators. These include classical, LLM-based, and diffusion-based approaches (CTGAN, GReaT, TabuLa, TapTap, PredLLM, TabSyn, TabDiff, and REaLTabFormer). All models are trained on identically preprocessed inputs to ensure methodological comparability.

Following synthetic data generation, all synthetic datasets undergo a validation and correction stage in which codebook constraints, ordinal and categorical validity rules, and REAL-anchored continuous bounds are enforced, with invalid records removed and correction effects logged prior to evaluation.

At the distributional level, marginal distributions, ordinal category fidelity, and covariate correlations are assessed to establish baseline statistical similarity. At the structural

level, SEM analyses were conducted simultaneously on real and synthetic data to extract global model fit indices, factor loadings, structural path coefficients, explained variance (R^2), reliability metrics, and discriminant validity statistics. Both PLS-SEM (primary) and CB-SEM (secondary) were employed to evaluate robustness across estimation paradigms.

In addition to utility-oriented evaluation, privacy preservation is assessed using established disclosure risk metrics, including distance-based and membership inference-oriented measures. All results are standardised and aggregated to facilitate systematic comparison across generators and datasets.

4.2 Datasets

4.2.1 PISA and TIMSS Dataset Background

This study uses data from the Programme for International Student Assessment (PISA) 2022, a large-scale international assessment designed to measure student achievement and contextual factors related to educational outcomes.

This study also draws on data from the Trends in International Mathematics and Science Study (TIMSS) 2023, a large-scale international assessment administered by the International Association for the Evaluation of Educational Achievement (IEA).

PISA is coordinated by the OECD, while TIMSS is administered by the International Association for the Evaluation of Educational Achievement (IEA).

4.3 Structural Coherence Evaluation Framework

A multi-layered Structural Equation Modelling (SEM)-based evaluation framework was designed and applied uniformly to real and synthetic datasets using R Programming. The framework integrates both variance-based SEM (PLS-SEM) and covariance-based SEM (CB-SEM), enabling complementary perspectives on measurement quality, explanatory power, and global model fit.

4.3.1 Pre-SEM Structural Checks

Before SEM estimation, composite scores for latent constructs were computed and used to construct a covariate correlation matrix.

4.3.2 PLS-SEM Estimation and PLSc Correction

Primary SEM estimation is conducted using Partial Least Squares Structural Equation Modelling (PLS-SEM) via the *SEMInR* package in R.

To correct for attenuation bias in reflective measurement models, the PLSc (consistent PLS) adjustment is then applied. This yields path coefficient estimates that are more comparable to covariance-based SEM.

All models are estimated using identical measurement and structural specifications for real and synthetic datasets. This ensures strict comparability.

4.3.3 Bootstrapping and Statistical Inference

Statistical inference is conducted making use of non-parametric bootstrapping with 1,000 resamples for PISA and 2000 resamples for TIMSS.

4.3.4 Structural Path Preservation

Structural coherence at the causal level is evaluated by comparing standardized path coefficients (β) between real and synthetic datasets. Standardization is performed using latent variable standard deviations to ensure scale-invariant comparison.

For each dataset, the following are reported:

- Direct effects
- Indirect and total effects
- Associated standard errors, t-statistics, and p-values

Preservation of sign, magnitude, and relative ordering of path coefficients is treated as evidence of structural coherence.

4.3.5 Explained Variance and Latent Relationships

Explanatory power is assessed using latent R^2 values for endogenous constructs. Both raw and adjusted R^2 values are extracted and compared across datasets.

In addition, latent correlation matrices among latent constructs are computed using estimated composite scores. These matrices are used to assess whether synthetic data preserve the internal dependency structure among key latent variables.

4.3.6 Measurement Quality and Discriminant Validity

Measurement-level coherence is evaluated using multiple complementary criteria, restricted to the primary latent constructs:

- **Internal consistency reliability:** Cronbach's α , composite reliability (pC), and pA
- **Convergent validity:** Average Variance Extracted (AVE)
- **Discriminant validity:** Fornell–Larcker criterion and HTMT ratios

Indicator loadings and indicator-level R^2 values are also extracted to assess factor loading similarity and measurement stability across datasets.

Control variables are intentionally excluded from measurement validity diagnostics, as they serve only structural roles.

4.3.7 CB-SEM Cross-Validation

To complement the variance-based analysis, the same conceptual model is estimated using covariance-based SEM (CB-SEM) via the lavaan package. CB-SEM estimation enables assessment of global model fit, including indices such as CFI, TLI, RMSEA, and SRMR.

CB-SEM outputs are used for:

- Global fit comparison between real and synthetic datasets
- Latent R^2 validation

This dual-SEM approach strengthens the evaluation by ensuring that conclusions are not artefacts of a single estimation paradigm.

4.4 Output Standardisation and Reproducibility

All computed metrics - spanning correlations, path coefficients, reliability indices, validity diagnostics, and fit measures are tracked. Consistent naming conventions and logging are employed to ensure traceability and reproducibility across experimental runs.

Detailed mathematical definitions and thresholds for all evaluation metrics are provided in Appendix A.

4.5 Mapping SEM Constructs to Machine Learning Training Objectives

A central challenge in this study arises from the mismatch between Structural Equation Modelling (SEM), which operates on latent constructs estimated from observed indicators, and synthetic data generators that require variable-centric inputs such as explicit target variables or conditioning columns. Because latent variables are not directly observed, aligning SEM structure with model training interfaces requires careful methodological design.

Composite scores for the structurally endogenous latent construct - representing the primary outcome variable in the SEM - were obtained from PLS-SEM estimation on the real dataset. These scores were incorporated into the training data only when required by specific generators. For models capable of unconditional generation (e.g., GReaT and Tabula), no target or conditioning column was provided.

Several alternative strategies were explored before adopting this configuration. Conditioning generators on SEM-derived latent composite scores was initially evaluated for models supporting conditional inputs; however, this approach did not produce stable or structurally coherent synthetic datasets suitable for downstream SEM analysis. For generators requiring supervised inputs (e.g., TabDiff, TabSyn, and PredLLM), principal component scores derived from observed indicators and arbitrary single-column targets

were also examined but failed to preserve the structural relationships encoded in the SEM.

The final approach uses the endogenous latent construct composite score as the target variable for generators requiring supervised training. This was obtained from SEM Model in R. This choice is theoretically justified as endogenous constructs in linear SEM function similarly to dependent variables in regression models. To preserve the integrity of SEM evaluation, the generated target column was discarded prior to analysis, and latent variables were re-estimated exclusively from observed indicators using standard SEM procedures.

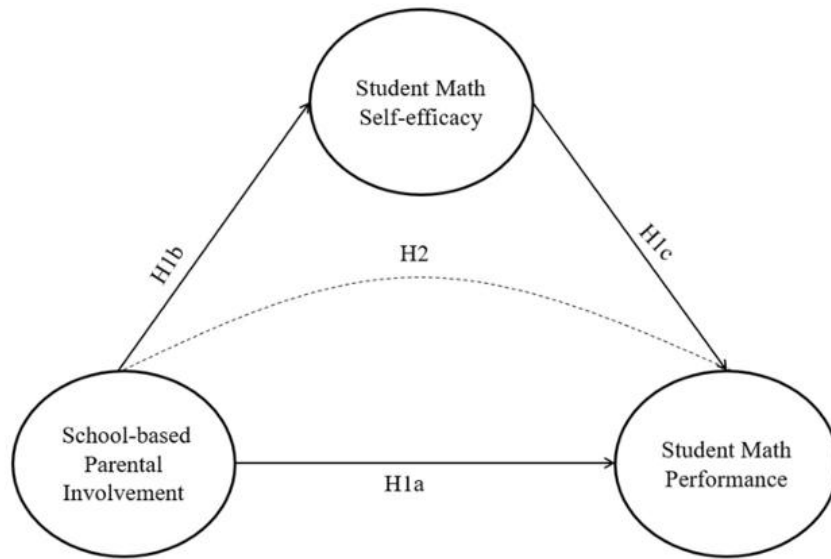
This alignment strategy enables heterogeneous generative techniques - ranging from unconditional LLM-based models to supervised diffusion-based approaches - to be benchmarked within a unified framework while maintaining the structural semantics of the SEM. Further details on this are provided in Appendix G.

5. Experiments and Results

5.1 Processing performed on PISA

Student-level and school-level datasets were obtained and processed using Python. For the primary analyses, the dataset was restricted to a single country (Singapore, SGP) to speed up experiments and results. While the analytical framework follows (Djekourmane et al., 2025), several additional design choices were adapted for the present study, including alternative missing-data handling, and preprocessing requirements for synthetic data generation.

Following the framework in (Djekourmane et al., 2025), a subset of variables was selected - student-level variables include demographic covariates (grade, gender, age, immigration status, maternal education, and socioeconomic status), indicators of student motivation (SMS), and mathematics achievement measured using ten plausible values (SMP). School-level variables include indicators of parental involvement (SPI) and basic school characteristics. The structural path diagram of the latent constructs discussed in (Djekourmane et al., 2025) is shown below.



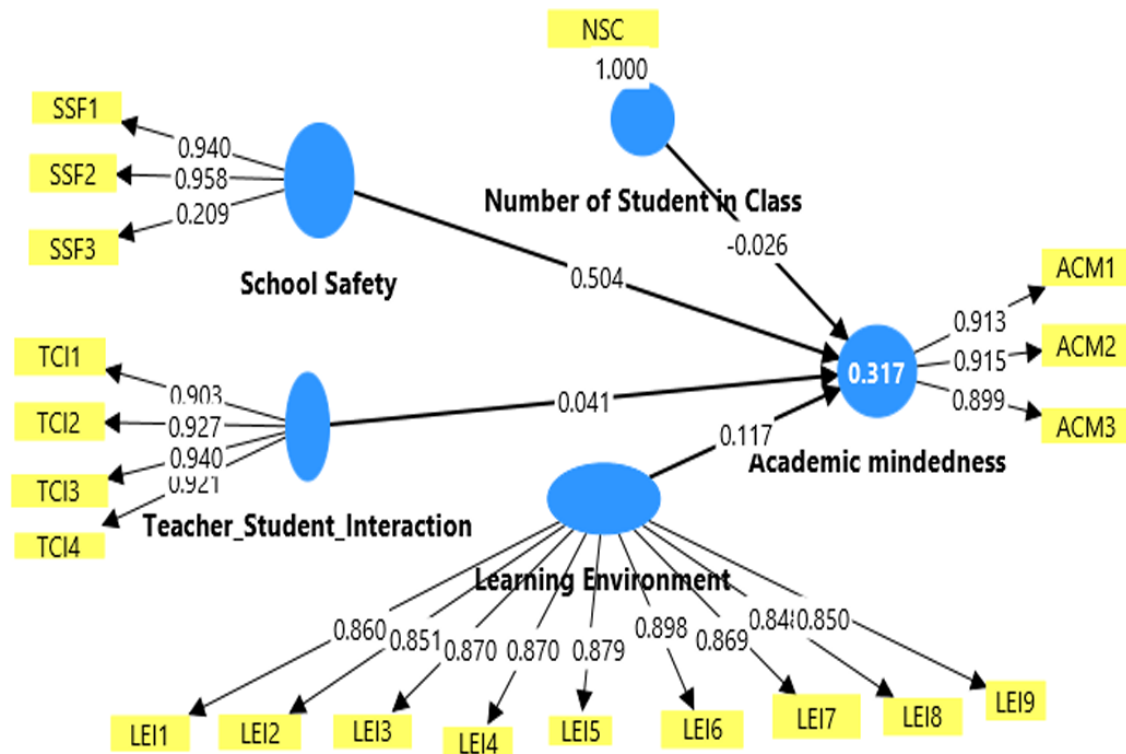
Source: (Djekourmane et al., 2025)

5.2 Processing done on TIMSS

For the present analysis, the dataset was restricted to a single country (Singapore, SGP) to speed up experiments and results.

The selection of constructs and the specification of the SEM were guided by the theory-driven framework proposed by (Olatunbosun & Pillay, 2025), who examined the role of supportive school climate factors in shaping learners' academic mindedness using TIMSS 2023 data. Consistent with this framework, the present study operationalises *academic mindedness* (ACM) as the primary endogenous construct, measured using student-reported indicators of engagement and effort. Exogenous latent constructs include *school safety and support* (SSF), *teacher-student interaction* (TCI), and *learning environment and instructional emphasis* (LEI), alongside a single-item school-level control capturing class size (NSC). These constructs reflect theoretically motivated dimensions of the school and instructional context hypothesised to influence students' academic dispositions.

The structural path diagram in (Olatunbosun & Pillay, 2025) is shown below:



Source: (Olatunbosun & Pillay, 2025)

5.3 Real and Synthetic Data Preprocessing

Real data preprocessing included variable type classification, listwise deletion for SEM indicators, simple imputation for non-indicator covariates, and removal of identifiers. The same cleaned dataset served as input to all generators.

Synthetic preprocessing involved some generators requiring minor model-specific transformations (e.g., standardisation or numeric-only inputs). These adjustments were applied only when required by architectural constraints and are detailed in Appendix C.

In addition, generator-specific requirements for categorical feature specification were addressed during preprocessing. For diffusion-based models that require explicit designation of categorical variables (e.g., TabDiff and TabSyn), all discrete variables were supplied as categorical features, including those with high cardinality. Empirically, this configuration substantially improved generation quality compared with restricting categorical designation to low-cardinality variables.

However, for LLM-based generators, high-cardinality discrete variables were not explicitly treated as categorical and were instead provided as regular variables. This approach produced more stable outputs. That is likely because transformer-based models can internally represent discrete values without requiring explicit categorical encoding. These model-specific input configurations were applied for satisfying architectural requirements whilst maintaining comparability across techniques.

5.4 Model Training and Fine-Tuning Strategy

All synthetic data generators evaluated in this study require model training or fine-tuning on the real dataset prior to sampling. To ensure comparability across methods, each generator was trained for optimal performance using optimal hyperparameters. Training and sampling were executed on the NeSI high-performance computing (HPC) platform or on Raapoi (Victoria University HPC Cluster) using Slurm-based job scheduling to ensure reproducibility and consistent resource allocation.

For LLM-based/Transformer-based generators (GReaT, PredLLM, TapTap, Tabula, and REaLTabFormer), training involved fine-tuning pretrained transformer models on the task-specific tabular dataset. In LLM-based cases, pretrained language models (DistilGPT-2 variants) served as initial weights, and models were fine-tuned using autoregressive objectives appropriate to each framework. Five synthetic datasets were generated from each trained model to capture sampling variability.

Diffusion-based generation was implemented using TabDiff and TabSyn, following the authors' recommended multi-stage workflow and minor patches applied to customize for this study (patches provided in Appendix B). Dataset preparation, model training, and sampling were executed as separate Slurm jobs. First, the processed dataset was converted into the TabDiff-specific format using a dedicated preprocessing script. Model training was then performed using the default diffusion architecture and a learnable noise schedule, without checkpoint reuse across runs. Finally, synthetic datasets were generated by repeatedly sampling from the trained diffusion model, producing multiple independent synthetic replicates of the same size as the real dataset.

Across all methods, no manual hyperparameter tuning was performed to optimise performance on structural metrics. Instead, models were trained using stable, method-recommended configurations to reflect realistic usage scenarios. Where applicable, model selection decisions (e.g. Tabula checkpoint choice) were guided by theory-relevant criteria (directional consistency and rank preservation) rather than training loss alone. This strategy ensures that observed differences in structural coherence arise from inherent modelling characteristics rather than differential optimisation effort.

5.5 Post-generation processing of synthetic data

All synthetic datasets derived from both PISA and TIMSS underwent post-generation validation and correction. Synthetic records were checked against codebook-defined category sets, admissible ordinal ranges, and continuous bounds anchored to the real data distribution. Rows with numerical values outside of official codebook limits or ± 1 standard deviation away from real data mean were tracked. Records violating any constraint were removed, with all corrections logged to enable auditability and systematic cross-technique comparison.

5.6 Training configuration, Evaluation and reporting metrics

In theory, the performance of synthetic data generators is assessed along three complementary criteria: distributional fidelity (statistical similarity to the real data), downstream analytical utility, and privacy protection. In this study, downstream utility is defined with respect to Structural Equation Modelling (SEM). Accordingly, evaluation focuses on SEM-specific criteria, including correlation preservation, measurement validity, structural coherence, and global model fit, in addition to distributional fidelity and privacy risk. Metrics corresponding to each criterion are computed and reported, and full mathematical definitions, thresholds and aggregation procedures are provided in Appendices D.

As this work is designed as a benchmarking study, model configurations are standardised to ensure fair and comparable evaluation across generators. Limited calibration of shared hyperparameters was performed to improve downstream SEM utility across all methods, while deliberately maintaining low implementation complexity to ensure accessibility for researchers in the SEM community without specialised AI or deep learning expertise. For LLM-based generators, training is conducted using a common backbone model (DistilGPT-2) and a consistent set of shared hyperparameters, namely batch size and number of epochs. These parameters were selected because they are uniformly exposed across all frameworks, whereas default settings were used for other parameters. Restricting tuning to shared parameters reduces implementation-specific advantages and improves fairness, reproducibility, and comparability of results. In some cases, parameters such as gradient accumulation steps were adjusted to accommodate the small sample size of the TIMSS dataset, ensuring that generators produced valid outputs suitable for downstream SEM evaluation.

Several synthetic data generators require a single target variable as part of a supervised or conditional learning setup. In such cases, the structurally endogenous latent variable from the SEM - representing the final-outcome construct - was used as the target column. The corresponding latent composite score was extracted from the PLS-SEM estimation on the real data and provided to the generator solely to satisfy training interface requirements. Importantly, while this construct is endogenous within the SEM, its role as a target variable is purely algorithmic and does not alter the structural interpretation of the model; furthermore, when SEM analyses were performed on the synthetic datasets, this generated target column was discarded, and latent variables were re-estimated exclusively from their observed indicators.

5.7 Reproducibility and Computational Environment

Experiments were executed on the New Zealand eScience Infrastructure or Rāpoi HPC cluster using NVIDIA A100 GPUs (4 CPUs and 64 GB RAM per task). Analyses were conducted using Python 3.11 and R with relevant packages. All experiments were

orchestrated using Slurm job scripts with isolated virtual environments and structured logging to ensure reproducibility.

Minor compatibility patches were applied to the REaLTabFormer library for ensuring stable execution with the installed HuggingFace Transformers version and for guaranteeing independent training across experimental runs.

Similarly, minor compatibility patches were also applied to the TabDiff and TapTap for supporting reproducible and methodologically consistent benchmarking.

All source-level changes are documented as patch files in Appendix B.

5.8 Results

5.1 Overview of Cross-Dataset Performance

This section reports the empirical performance of synthetic data generators on the TIMSS 2023 and PISA 2022 datasets using the unified SEM-based evaluation framework. For each dataset, results are summarized using consolidated metric tables, which aggregate normalized component scores across distribution fidelity, structural fidelity, measurement fidelity, validity, global fit, and privacy. Detailed metric breakdowns, granular diagnostics, and replication results are provided in Appendices D and E.

Across both datasets, results reveal clear and stable performance among techniques, with a small subset of generators outperforming others across multiple evaluation dimensions. Even though absolute rankings differ slightly between TIMSS and PISA, the relative strengths and weaknesses of each technique remain largely consistent.

5.1 TIMSS 2023 (Singapore): Summary Results

Table 1

Detailed Metric Breakdown by Technique (TIMSS 2023, Singapore)

Metric	TabDiff	TabSyn	GReaT	PredLLM	TapTap	REaLTabFormer	CTGAN	Tabula
— DISTRIBUTION FIDELITY —								
Mean $ \Delta\mu \downarrow$	0.030	0.080	0.110	0.080	0.060	0.300	0.250	1.330
Mean $ \Delta\sigma \downarrow$	0.030	0.050	0.010	0.030	0.120	0.050	0.030	0.300
Continuous adherence $\uparrow$	1.000	1.000	1.000	0.852	0.994	0.923	1.000	0.758
Category adherence $\uparrow$	1.000	1.000	1.000	0.000	0.000	0.000	1.000	1.000
Composite error $\downarrow$	0.060	0.130	0.120	1.258	1.186	1.427	0.280	1.872

Distribution composite $\uparrow$	0.997	0.961	1.000	0.436	0.535	0.455	0.940	0.000
Distribution rank	2	3	1	7	5	6	4	8
— STRUCTURAL FIDELITY —								
Directional consistency $\uparrow$	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.750
Rank preservation (Spearman ρ) $\uparrow$	1.000	1.000	1.000	0.600	0.800	0.400	-0.200	1.000
Latent corr $ \Delta r \downarrow$	0.057	0.030	0.058	0.058	0.120	0.180	0.216	0.235
Covariate $ \Delta r \downarrow$	0.020	0.021	0.053	0.040	0.071	0.044	0.061	0.162
Structural composite $\uparrow$	0.717	0.748	0.659	0.625	0.521	0.274	0.283	0.083
Structural rank	2	1	3	4	5	7	6	8
— MEASUREMENT FIDELITY —								
Loading MAD $\downarrow$	0.097	0.016	0.026	0.014	0.092	0.289	0.259	0.378
Reliability MAD $\downarrow$	0.508	0.371	0.177	0.429	0.336	0.174	1.058	1.131
Measurement composite $\uparrow$	0.711	0.895	0.983	0.867	0.809	0.623	0.202	0.000
Measurement rank	5	2	1	3	4	6	7	8
— VALIDITY —								
HTMT RMSE $\downarrow$	0.035	0.040	0.070	0.069	0.141	0.091	0.172	0.070
CFI $\uparrow$	0.960	0.910	0.930	0.960	0.930	0.820	0.920	0.920
TLI $\uparrow$	0.950	0.890	0.910	0.950	0.920	0.790	0.910	0.910
RMSEA $\downarrow$	0.040	0.070	0.070	0.050	0.070	0.110	0.060	0.060
SRMR $\downarrow$	0.050	0.060	0.060	0.050	0.060	0.090	0.060	0.070
Validity composite $\uparrow$	1.000	0.805	0.729	0.858	0.478	0.296	0.366	0.707
Validity rank	1	3	4	2	6	8	7	5
— PRIVACY —								
Exact Match Rate $\downarrow$	0.000	0.143	0.065	0.000	0.000	0.000	0.000	0.000
NNDR $\uparrow$	2.623	1.277	1.622	2.310	2.172	2.334	2.832	4.046
Membership inference risk $\downarrow$	0.762	0.891	0.937	0.848	0.856	0.840	0.751	0.442
DCR p05 $\uparrow$	1.742	0.000	0.283	1.463	1.277	1.300	1.861	1.755
Privacy composite $\uparrow$	0.697	0.000	0.189	0.583	0.540	0.567	0.739	1.000
Privacy rank	3	8	7	4	6	5	2	1
— OVERALL —								
Composite score $\uparrow$	1.000	0.833	0.827	0.795	0.537	0.180	0.166	0.000
Overall rank	1	2	3	4	5	6	7	8

Note. Rows represent individual evaluation metrics. Errors are shown as positive magnitudes. Normalized scores are scaled to [0,1], where higher is better. Overall composite score is computed as a weighted sum of normalized component scores with the following weights: Distribution: 0.13; Path: 0.14; Latent corr: 0.08; Covariate: 0.08; Loading: 0.12; Reliability: 0.10; HTMT: 0.12; Fit: 0.09; Privacy: 0.14. All component scores are min-max normalized to [0,1] prior to aggregation. Detailed metric-level explanation and diagnostics supporting these summaries are provided in Appendices D and E.

Table 1 reports the consolidated evaluation results for the TIMSS dataset. On an overall basis, TabDiff achieves the highest composite score and ranks first among all techniques. TabSyn and GReaT are close followers. These methods perform best on distributional, structural and measurement metrics.

From a distributional perspective, TabDiff and GReaT show near-perfect continuous and categorical adherence. They also show minimal deviations in means and variances, and correspondingly high distribution composite scores. In contrast, Tabula and

REaLTabFormer exhibit substantial distributional distortions which is reflected in large mean and variance deviations and low composite rankings.

In terms of structural fidelity, TabSyn and TabDiff achieve perfect directional consistency and strong rank preservation of structural paths versus the real data. Latent and covariate correlation errors are lowest for these techniques which indicates effective preservation of relational structure. CTGAN and REaLTabFormer, show weaker structural alignment.

For measurement fidelity, GReaT ranks first, followed by TabSyn and PredLLM. This reflects low loading and reliability deviations for all latent constructs. Techniques with weaker distributional performance also tend to exhibit larger measurement distortions. This suggests compounding errors in all stages of the SEM pipeline.

Validity and global fit diagnostics also reinforce these patterns. TabDiff achieves the highest validity composite score, driven by low HTMT error and strong CB-SEM fit indices. Several techniques achieve acceptable global fit thresholds, but only a subset maintain consistent discriminant validity relative to the real data.

Finally, privacy results show a clear trade-off with fidelity. Tabula and CTGAN achieve stronger privacy protection but at the cost of substantial fidelity loss. TabDiff and TabSyn balance moderate privacy protection against high analytical utility.

Overall, the TIMSS results suggest diffusion-based and transformer-based generators provide the most reliable synthetic substitutes for SEM-based educational analysis. Adversarial and autoencoder-based approaches struggle to jointly satisfy fidelity and privacy constraints.

5.1 PISA 2022 (Singapore): Summary Results

Table 2

Detailed Metric Breakdown by Technique (PISA 2022)

Metric	TabSyn	TabDiff	TapTap	GReaT	PredLLM	Tabula	CTGAN	REaLTabFormer
— DISTRIBUTION FIDELITY —								
Mean $ \Delta\mu \downarrow$	0.030	0.030	0.040	0.180	0.040	0.170	0.100	0.240
Mean $ \Delta\sigma \downarrow$	0.070	0.240	0.030	0.250	0.050	0.260	0.140	0.090
Continuous adherence $\uparrow$	1.000	1.000	0.989	1.000	0.999	1.000	1.000	0.998
Category adherence $\uparrow$	1.000	1.000	0.500	1.000	0.500	1.000	0.768	0.500
Composite error $\downarrow$	0.100	0.270	0.581	0.430	0.591	0.430	0.472	0.832
Distribution composite $\uparrow$	1.000	0.674	0.174	0.340	0.537	0.342	0.514	0.000
Distribution rank	1	2	7	6	3	5	4	8
— STRUCTURAL FIDELITY —								
Directional consistency $\uparrow$	1.000	1.000	0.842	0.842	0.842	0.789	0.842	0.789
Rank preservation (Spearman ρ) $\uparrow$	0.872	0.707	0.626	0.675	0.609	0.825	0.121	0.447
Latent corr $ \Delta r \downarrow$	0.029	0.017	0.033	0.030	0.029	0.039	0.044	0.086
Covariate $ \Delta r \downarrow$	0.025	0.016	0.033	0.044	0.043	0.034	0.031	0.057
Indirect $ \Delta\beta \downarrow$	0.022	0.013	0.011	0.013	0.010	0.013	0.006	0.030

Metric	TabSyn	TabDiff	TapTap	GReaT	PredLLM	Tabula	CTGAN	REaLTabFormer
Structural composite ↑	0.736	0.896	0.631	0.567	0.590	0.586	0.557	0.026
Structural rank	2	1	3	6	4	5	7	8
—MEASUREMENT FIDELITY—								
Loading MAD ↓	0.028	0.111	0.046	0.037	0.073	0.037	0.084	0.108
Reliability MAD ↓	0.032	0.074	0.017	0.030	0.033	0.033	0.059	0.055
Measurement composite ↑	0.870	0.000	0.893	0.832	0.592	0.803	0.294	0.185
Measurement rank	2	8	1	3	5	4	6	7
— VALIDITY —								
HTMT RMSE ↓	0.040	0.028	0.046	0.024	0.045	0.059	0.069	0.080
CFI ↑	0.880	0.720	0.930	0.850	0.910	0.890	0.880	0.840
TLI ↑	0.870	0.680	0.920	0.830	0.900	0.870	0.870	0.820
RMSEA ↓	0.100	0.090	0.080	0.100	0.080	0.090	0.070	0.110
SRMR ↓	0.090	0.080	0.080	0.090	0.070	0.090	0.080	0.100
Validity composite ↑	0.560	0.467	0.772	0.639	0.812	0.452	0.526	0.000
Validity rank	4	6	2	3	1	7	5	8
— PRIVACY —								
Exact Match Rate ↓	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
NNDR ↑	99.200	271.228	89.826	80.810	97.338	83.650	187.354	110.220
Membership inference risk ↓	0.863	0.383	0.917	0.964	0.901	0.944	0.531	0.832
DCR p05 ↑	56.436	146.862	57.721	47.822	60.451	55.591	104.322	63.969
Privacy composite ↑	0.119	1.000	0.076	0.000	0.108	0.043	0.625	0.182
Privacy rank	4	1	6	8	5	7	2	3
— OVERALL —								
Composite score ↑	1.000	0.884	0.824	0.800	0.748	0.702	0.598	0.000
Overall rank	1	2	3	4	5	6	7	8

Note. Rows represent individual evaluation metrics. Errors are shown as positive magnitudes. Normalized scores are scaled to [0,1], where higher is better. Overall composite score is computed as a weighted sum of normalized component scores with the following weights: Distribution: 0.13; Path: 0.13; Latent corr: 0.08; Covariate: 0.08; Loading: 0.12; Reliability: 0.10; HTMT: 0.12; Fit: 0.09; Privacy: 0.15; Indirect: 0.08. All component scores are min-max normalized to [0,1] prior to aggregation. Detailed metric-level explanation and diagnostics supporting these summaries are provided in Appendices D and E.

Table 2 presents the evaluation for the PISA dataset. TabSyn ranks first on an overall basis here. This is followed by TabDiff and TapTap even though the same group of high-performing techniques remains evident.

Distribution fidelity is dominated by TabSyn and TabDiff. Both of these achieve minimal deviations in means and variances and perfectly adhere to observed data bounds. Category adherence issues are more pronounced in PISA than in TIMSS. This is particularly true for techniques that generate continuous approximations of ordinal variables. These effects are reflected in lower distribution composite scores for several LLM-based models.

In structural metrics, TabDiff ranks first. It achieves the highest structural composite score from strong rank preservation, low correlation error, and accurate indirect effects. TabSyn is second here and REaLTabFormer again shows worst performance.

Measurement fidelity results are similar to TIMSS. TapTap and TabSyn achieve the highest composite scores and lowest loading and reliability deviations. Techniques that underperform distributionally exhibit larger measurement errors which shows there is interdependence between marginal realism and psychometric stability.

For validity and global fit, PredLLM has the highest validity composite score from strong HTMT performance and acceptable CB-SEM fit indices. However, this advantage does not translate into superior overall ranking due to weaker distributional and structural performance.

Privacy outcomes differ markedly from TIMSS. TabDiff achieves the highest privacy composite score, followed by CTGAN, reflecting very large nearest-neighbour distances and low membership inference risk. Nevertheless, high privacy performance is again associated with reduced analytical fidelity for several techniques.

Taken together, the PISA results confirm that no single technique dominates all evaluation dimensions, but TabSyn and TabDiff consistently offer the most balanced trade-off between fidelity, validity, and privacy.

5.4 Cross-Dataset Comparison and Key Findings

Comparing results across TIMSS and PISA reveals three robust findings.

First, Diffusion based models - TabDiff and TabSyn rank the top two techniques on both datasets. This shows generalization in different educational contexts, measurement models, and structural specifications.

Second, distributional accuracy alone is insufficient: techniques with acceptable marginal distributions but weak structural or measurement fidelity fail to achieve high overall rankings, underscoring the importance of SEM-aware evaluation.

Third, privacy–utility trade-offs are systematic rather than incidental. Methods achieving maximal privacy protection do so by disrupting higher-order structure, limiting their suitability for substantive SEM-based inference.

5.5 How do we improve performance of LLM-Based Generators

Table 3

Effect of Model Capacity and Hyperparameter Tuning on LLM-Based Generators

Metric	TabDiff	GReaTLrg	TabSyn	GReaT	PredLLM	TapTap	REaLTabFrmr	CTGAN	Tabula
— DISTRIBUTION FIDELITY —									
Mean $ \Delta\mu \downarrow$	0.030	0.100	0.080	0.110	0.080	0.060	0.300	0.250	1.030
Mean $ \Delta\sigma \downarrow$	0.030	0.020	0.050	0.010	0.030	0.120	0.050	0.030	0.480

Metric	TabDiff	GReaTLrg	TabSyn	GReaT	PredLLM	TapTap	REaLTabFrmr	CTGAN	Tabula
Continuous adherence ↑	1.000	1.000	1.000	1.000	0.852	0.994	0.923	1.000	0.867
Category adherence ↑	1.000	1.000	1.000	1.000	0.000	0.000	0.000	1.000	1.000
Composite error ↓	0.060	0.120	0.130	0.120	1.258	1.186	1.427	0.280	1.643
Distribution composite ↑	1.000	0.983	0.968	0.987	0.282	0.558	0.358	0.923	0.000
Distribution rank	1	3	4	2	8	6	7	5	9
— STRUCTURAL FIDELITY —									
Directional consistency ↑	1.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	0.500
Rank preservation (Spearman ρ) ↑	1.000	0.600	1.000	1.000	0.600	0.800	0.400	-0.200	1.000
Latent corr Δr ↓	0.057	0.081	0.030	0.058	0.058	0.120	0.180	0.216	0.195
Covariate Δr ↓	0.020	0.068	0.021	0.053	0.040	0.071	0.044	0.061	0.148
Structural composite ↑	0.713	0.504	0.748	0.649	0.590	0.487	0.251	0.169	0.028
Structural rank	2	5	1	3	4	6	7	8	9
— MEASUREMENT FIDELITY —									
Loading MAD ↓	0.097	0.022	0.016	0.026	0.014	0.092	0.289	0.259	0.423
Reliability MAD ↓	0.508	0.151	0.371	0.177	0.429	0.336	0.174	1.058	1.236
Measurement composite ↑	0.734	0.991	0.897	0.974	0.872	0.820	0.653	0.283	0.000
Measurement rank	6	1	3	2	4	5	7	8	9
— VALIDITY —									
HTMT RMSE ↓	0.035	0.095	0.040	0.070	0.069	0.141	0.091	0.172	0.068
CFI ↑	0.960	0.950	0.910	0.930	0.960	0.930	0.820	0.920	0.820
TLI ↑	0.950	0.940	0.890	0.910	0.950	0.920	0.790	0.910	0.790
RMSEA ↓	0.040	0.060	0.070	0.070	0.050	0.070	0.110	0.060	0.080
SRMR ↓	0.050	0.050	0.060	0.060	0.050	0.060	0.090	0.060	0.080
Validity composite ↑	1.000	0.729	0.805	0.729	0.858	0.478	0.296	0.366	0.464
Validity rank	1	5	3	4	2	6	9	8	7
— PRIVACY —									
Exact Match Rate ↓	0.000	0.003	0.143	0.065	0.000	0.000	0.000	0.000	0.019
NNDR ↑	2.623	2.115	1.277	1.622	2.310	2.172	2.334	2.832	3.181
Membership inference risk ↓	0.762	0.825	0.891	0.937	0.848	0.856	0.840	0.751	0.691

Metric	TabDiff	GReaTLrg	TabSyn	GReaT	PredLLM	TapTap	REaLTabFrmr	CTGAN	Tabula
DCR p05 ↑	1.742	1.353	0.000	0.283	1.463	1.277	1.300	1.861	1.001
Privacy composite ↑	0.935	0.713	0.000	0.204	0.739	0.679	0.727	1.000	0.950
Privacy rank	3	6	9	8	4	7	5	1	2
— OVERALL —									
Composite score ↑	1.000	0.826	0.820	0.800	0.776	0.573	0.318	0.284	0.000
Overall rank	1	2	3	4	5	6	7	8	9

Note. Rows represent individual evaluation metrics. Errors are shown as positive magnitudes. Normalized scores are scaled to [0,1], where higher is better. Overall composite score is computed as a weighted sum of normalized component scores with the following weights: Distribution: 0.13; Path: 0.14; Latent corr: 0.08; Covariate: 0.08; Loading: 0.12; Reliability: 0.10; HTMT: 0.12; Fit: 0.09; Privacy: 0.14. All component scores are min–max normalized to [0,1] prior to aggregation. Detailed metric-level explanation and diagnostics supporting these summaries are provided in Appendices D and E.

To assess whether the comparatively weaker performance of LLM-based generators is intrinsic to the modelling paradigm or contingent on model capacity and tuning depth, an additional controlled evaluation was conducted using GReaT with GPT2Large model (denoted GReaT-Lrg). Table 3 reports the consolidated results alongside previously evaluated techniques.

Despite diffusion-based methods continuing to achieve the strongest overall performance, increasing LLM model capacity yields substantial improvements across multiple evaluation dimensions. GReaT-Lrg shows significantly improved distribution fidelity, achieving near-perfect adherence to marginal distributions and categorical constraints. It hence ranks second overall on the distribution composite.

Gains are specifically pronounced in measurement fidelity. That is where GReaT-Lrg achieves the highest measurement composite score among all techniques. This reflects substantially reduced loading and reliability deviations relative to the base GReaT model. Structural fidelity also improves relative to the baseline LLM configuration. Although, diffusion-based generators retain an advantage in rank preservation and correlation accuracy.

Validity and global fit indices for GReaT-Lrg remain competitive with leading diffusion-based approaches. This indicates that increased model capacity does not compromise psychometric plausibility. Privacy performance was moderate which demonstrates the trade-off between fidelity and disclosure risk.

Overall, these results indicate that the inferior performance of baseline LLM-based generators should not be interpreted as a limitation of the model type. Model capacity and hyperparameter tuning are critical for performance.

6. Discussion and Conclusion

6.1 Why Diffusion-Based Generators Consistently Perform Best

For both TIMSS 2023 and PISA 2022, diffusion-based generators (TabDiff and TabSyn) ranked the best in all the techniques. They achieve strong scores across distributional fidelity, structural alignment, measurement quality, and model validity. This dominance is not merely empirical but reflects fundamental differences in how diffusion models learn data structure. By denoising from noise to data diffusion models approximate the full joint distribution (Ho et al., 2020; Kotelnikov et al., 2023). This global modelling capability turned out to be well suited for SEM-based analysis. In SEM, preserving higher-order dependencies in latent constructs, indicators, and covariates stand of high importance (Bollen, 1989; Kline, 2023).

6.2 Interdependence Between Distributional, Measurement, and Structural Fidelity

In both datasets, the tight coupling between distributional accuracy, measurement fidelity, and structural validity turned out to be an important finding. Techniques exhibiting substantial deviations in marginal distributions or category adherence also tend to show inflated loading and reliability errors, followed by degraded structural path preservation. These patterns indicate that errors propagate through the SEM pipeline: marginal distortions compromise measurement quality, which in turn undermines structural inference—an effect long recognised in SEM theory (Bollen, 1989; Kline, 2016).

This interdependence is why SEM-aware evaluation is important. Assessing synthetic data quality using isolated metrics can give a misleading impression of analytical suitability. High-fidelity SEM inference requires consistency across all stages of the modelling pipeline consistent with recent critiques of marginal-only synthetic data evaluation (Pathare et al., 2023; Snokke et al., 2018).

6.3 Privacy–Utility Trade-offs Are Structural Rather Than Accidental

Privacy results across both datasets reveal a systematic trade-off between disclosure protection and analytical fidelity. Techniques achieving strong privacy metrics - such as high nearest-neighbour distances and low membership inference risk - typically do so by disrupting higher-order structure, leading to degraded measurement and structural performance. Conversely, high-fidelity generators necessarily retain closer proximity to the real data manifold, limiting the degree of privacy protection achievable without explicit privacy mechanisms.

6.4 Model Capacity and the Role of LLM-Based Generators

The additional capacity experiment with GReaT GPT2Large demonstrates that the comparatively weaker performance of baseline LLM-based generators should not be interpreted as a fundamental limitation of the paradigm. Increasing model capacity and tuning depth yields substantial improvements in distributional fidelity, measurement quality, and structural alignment, narrowing the performance gap with diffusion-based approaches.

6.7 Limitations and Scope

This study focused on using PLS SEM as a primary evaluation method while using CB SEM as a secondary evaluation. Using CB SEM as a primary evaluation method offers even more advantages and using Multi Group CFA technique for evaluation can be considered an even stricter evaluation. However, the study offers as a starting point for this type of research as because one of the earliest studies to explore SEM data generation.

6.8 Practical Trade-offs Between Diffusion and LLM-Based Generators

An important practical distinction between diffusion-based and LLM-based generators concerns the effort required to obtain usable synthetic data for SEM analysis. Diffusion-based methods such as TabDiff and TabSyn provide a comparatively *quick and robust solution*, requiring minimal hyperparameter tuning to achieve strong distributional and structural fidelity. This makes them attractive default choices in applied settings where rapid generation is prioritised or extensive tuning is infeasible.

However, this convenience comes with limitations, particularly in case of small-samples. Diffusion-based generator (specifically TabSyn) is optimised for moderate to large tabular datasets and does not explicitly account for extreme low-N settings. In the TIMSS subset analysed in this study ($N = 341$), TabSyn exhibits a markedly elevated exact match rate (0.14), indicating privacy leakage. This behaviour is consistent with overfitting to the training data, where repeated denoising steps effectively memorise rare records rather than learning a generalisable data manifold. As a result, diffusion models may fail badly on privacy metrics when applied naively to small datasets, despite otherwise strong fidelity performance.

By contrast, LLM-based generators demonstrate greater flexibility across evaluation dimensions when tuned appropriately, particularly in measurement fidelity, where they consistently achieve lower loading and reliability deviations than diffusion-based methods. Unlike diffusion models, LLMs benefit substantially from careful optimisation strategies, including gradient accumulation to stabilise training on small datasets. In preliminary experiments, failing to use gradient accumulation frequently resulted in unstable training dynamics, leading to invalid outputs or missing values in generated samples (specifically in case of TapTap). Proper accumulation effectively increases the

batch size without exceeding memory constraints, which is critical for small-N tabular synthesis.

Additionally, restricting LLM training to a narrow or default hyperparameter configuration produces suboptimal results. This can severely degrade privacy performance. Warmup scheduling plays a particularly important role in this context: insufficient warmup can cause early overfitting to limited training data, while excessive learning rates exacerbate memorisation.

Taken together, these findings suggest that diffusion-based generators offer a strong *out-of-the-box* solution for SEM-oriented synthetic data generation, but their robustness diminishes in small-sample privacy-sensitive settings. LLM-based generators, while more demanding in terms of tuning and computational care, provide greater adaptability and measurement stability when appropriately configured. Consequently, generator selection should be informed not only by aggregate performance rankings but also by dataset size, privacy requirements, and the resources available for model optimisation.

6.9 Conclusion

This study evaluated diffusion-based and LLM-based tabular generators for SEM-oriented synthetic data analysis across TIMSS 2023 and PISA 2022. Diffusion models provided the strongest out-of-the-box structural and distributional fidelity, while well-tuned LLM-based generators demonstrated competitive and often superior measurement stability. Results highlight the interdependence of marginal realism, psychometric validity, and structural coherence, and reveal systematic privacy–utility trade-offs, particularly in small-sample settings. Overall, generator suitability depends on dataset size, tuning depth, and analytical objectives. The proposed SEM-aware evaluation framework offers a structured approach for assessing synthetic data reliability in educational and behavioural research contexts.

Use of Generative AI Tools

Portions of this manuscript were drafted with the assistance of generative AI tools to improve clarity and language expression. All technical content, analysis, experimental design, code implementation, and interpretation of results were conducted independently by the author. The author reviewed, edited, and verified all generated text to ensure accuracy and alignment with the underlying research.

7. Appendices

7.1 Appendix A – SEM Metric Definitions

1. Measurement-Level Integrity

Measurement models assess whether observed indicators reliably represent latent constructs.

1.1 Factor Loadings

Definition: Strength of the relationship between an observed indicator and its latent construct.

Indicator reliability:

$$\lambda_i = \text{loading of indicator } i$$

Recommended:

$$|\lambda_i| \geq 0.70$$

Interpretation:

- High loadings → indicators strongly reflect the construct
- Low/drifted loadings → construct meaning changes
- Cross-loadings → loss of discriminant validity
- In synthetic data: large deviations indicate distortion of measurement structure

(Hair, 2014)

1.2 Internal Consistency (Cronbach's α)

Definition: Degree to which indicators of a construct measure the same underlying concept.

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum_{i=1}^k \sigma_i^2}{\sigma_T^2} \right)$$

where

k = number of indicators,

σ_T^2 = total variance.

Threshold:

$$\alpha \geq 0.70$$

Interpretation:

- High → indicators behave cohesively
- Low → noisy or incoherent measurement
- Synthetic drop → reliability degradation

(Cronbach, 1951)

1.3 Composite Reliability (ρ_C)

Definition: Composite reliability quantifies the internal consistency of a latent construct by incorporating the magnitude of individual factor loadings, thereby accounting for unequal indicator contributions.

$$\rho_C = \frac{(\sum \lambda_i)^2}{(\sum \lambda_i)^2 + \sum (1 - \lambda_i^2)}$$

Preferred in SEM due to unequal loadings:

$$\rho_C \geq 0.70$$

Interpretation:

- Higher → stronger internal consistency and more reliable measurement
- Lower → greater measurement error and weaker construct representation

(Fornell & Larcker, 1981)

1.4 Reliability A (ρ_A)

Definition:

Reliability A is a consistent reliability estimator that corrects bias in composite reliability and provides a stable estimate of true score reliability in SEM.

$$\rho_A = \frac{(\sum \lambda_i)^2}{(\sum \lambda_i)^2 + \sum \theta_i}$$

where θ_i are error variances.

Interpretation:

- Higher → more dependable and stable construct measurement
- Lower → increased unreliability and inconsistency among indicators

(Dijkstra & Henseler, 2015)

1.5 Average Variance Extracted (AVE)

Definition:

Average variance extracted measures the proportion of variance in indicators that is explained by the latent construct relative to measurement error, reflecting convergent validity.

$$AVE = \frac{\sum \lambda_i^2}{k}$$

Criterion:

$$AVE \geq 0.50$$

Interpretation:

- Higher → indicators share substantial common variance with the construct
- Lower → error variance dominates, indicating weak convergent validity

(Fornell & Larcker, 1981)

1.6 Discriminant Validity

(a) Fornell–Larcker criterion

Definition:

Assesses whether a construct shares more variance with its own indicators than with other constructs.

$$\sqrt{AVE_j} > r_{jk} \forall k \neq j$$

Interpretation:

- Larger separation → constructs are distinct
- Smaller separation → constructs overlap conceptually

(b) HTMT

Definition:

Ratio of between-construct correlations to within-construct correlations; a more sensitive test of discriminant validity.

$$HTMT = \frac{\text{mean heterotrait correlations}}{\text{mean monotrait correlations}}$$

Threshold:

$$HTMT < 0.85 \text{ or } 0.90$$

Interpretation:

- **Lower (< 0.85–0.90):** constructs clearly distinct
- **Higher:** constructs may measure the same concept

(Henseler et al., 2015)

2. Structural-Level Preservation

Evaluates whether latent relationships and theoretical pathways remain intact.

2.1 Path coefficients

Definition:

Standardised regression weights representing the strength and direction of relationships between latent constructs.

$$\beta_{jk}$$

Compare synthetic vs real:

$$\Delta\beta = |\beta_{synthetic} - \beta_{real}|$$

Interpretation:

- Larger magnitude → stronger effect
- Smaller magnitude → weaker or negligible effect
- Positive/negative sign → direction of influence

2.2 Explained variance

Definition:

Proportion of variance in an endogenous construct explained by its predictors.

$$R^2 = 1 - \frac{\sum(y - \hat{y})^2}{\sum(y - \bar{y})^2}$$

Interpretation:

- Higher → stronger explanatory or predictive power
- Lower → limited explanatory capacity

(Kline, 2023)

2.3 Mediation / indirect effects

Definition: Effect of one construct on another transmitted through an intermediate construct.

$$\beta_{indirect} = \beta_a \times \beta_b$$

Interpretation:

- Larger magnitude → stronger mediated influence
- Smaller magnitude → weaker or negligible mediation

3. Global Model Fit (Covariance Reproduction)

Evaluates whether the model reproduces observed covariance structure.

3.1 Comparative Fit Index (CFI)

Definition: Incremental fit index comparing the hypothesised model with a null (independence) model.

$$CFI = 1 - \frac{\chi^2_{model} - df_{model}}{\chi^2_{baseline} - df_{baseline}}$$

Interpretation:

- Higher → better model fit
- Lower → poorer fit relative to baseline

3.2 Tucker–Lewis Index (TLI)**Definition:**

Fit index adjusting model fit for complexity (degrees of freedom).

$$TLI = \frac{(\chi^2_{baseline}/df_{baseline}) - (\chi^2_{model}/df_{model})}{(\chi^2_{baseline}/df_{baseline}) - 1}$$

Interpretation:

- Higher → better fit with appropriate parsimony
- Lower → inadequate fit or overly complex model

3.3 RMSEA

Definition:

Measures discrepancy between observed and model-implied covariance matrices per degree of freedom.

$$RMSEA = \sqrt{\frac{\chi^2 - df}{df(N - 1)}}$$

Interpretation:

- Lower → closer model approximation
- Higher → greater model misfit

3.4 SRMR

Definition:

Average standardized difference between observed and model-implied correlations.

$$SRMR = \sqrt{\frac{1}{p(p+1)/2} \sum (r_{ij}^{obs} - r_{ij}^{model})^2}$$

Interpretation:

- Lower → smaller residual errors
- Higher → larger discrepancies

Cutoffs (approximate):

- CFI/TLI ≥ 0.90–0.95
- RMSEA ≤ 0.06–0.08
- SRMR ≤ 0.08

(Hu & Bentler, 1999)

7.2 Appendix B — Library Patches

The actual patches have been attached with this submission.

Below is the summary of the patches.

B.1 REaLTabFormer Modifications

Library	version:	realtabformer	(installed	via	pip)
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Purpose: compatibility with Transformers $\geq 4.x$ and deterministic independent training

File	Modification	Rationale
realtabformer.py	eval_strategy → evaluation_strategy	API compatibility
rtf_trainer.py	removed processing_class=tokenizer	deprecated argument
realtabformer.py	resume_from_checkpoint=False	prevent state carryover between replicates

Reproduction Commands

```
sed -i 's/\beval_strategy\b/evaluation_strategy/g' realtabformer.py
```

```
sed -i '/processing_class=tokenizer/d' rtf_trainer.py
```

```
sed -i 's/resume_from_checkpoint=True/resume_from_checkpoint=False/'
realtabformer.py
```

B.2 TabDiff Trainer Modifications

Library: TabDiff

File: modified:

trainer.py

Purpose: deterministic sampling, replicate isolation, centralised evaluation

Summary of changes

Change	Rationale
Added TABDIFF_SAMPLE_TAG path control	isolate replicate outputs
Removed internal metric computation	prevent inconsistent evaluation
Converted evaluation routine to sampling-only	enforce unified external evaluation

B.3 TabDiff Main Script Modifications

File **modified:** **main.py**
Purpose: disable library-internal evaluation to enforce unified external evaluation
Summary

Change	Rationale
Disabled <code>trainer.report_test()</code>	prevent internal density/MLE/C2ST metrics
Centralised evaluation externally	ensure identical evaluation across generators

B.4 TapTap Dataset Modifications

Library: **TapTap**
File **modified:** **taptap_dataset.py**
Purpose: HuggingFace datasets / Trainer compatibility, correct batched access, and reproducible HPC execution
Summary of changes

Change	Rationale
Revised <code>_getitem</code> to support int, list, and slice indexing	ensure HuggingFace-compliant batched access
Returned dictionary-of-lists for batched queries	required by datasets and Trainer APIs
Enforced tokenizer-level truncation (<code>truncation=True, max_length</code>)	avoid implicit token slicing
Guaranteed minimum sequence length for causal LM	prevent invalid loss computation
Removed reliance on <code>BatchEncoding</code> objects	ensure deterministic collation under gradient accumulation

Scope **clarification**
These modifications address data-loading and interface-level inconsistencies only and do not alter TapTap’s modelling assumptions, text-based tabular representation, or training objective. They were required to ensure stable and reproducible training behaviour under gradient accumulation and multi-replica HPC execution.

7.3 Appendix C – Synthetic Data Preprocessing

In addition to the shared preprocessing pipeline, several generators required model-specific data transformations dictated by architectural constraints rather than analytical preference.

For GReaT, the merged dataset was used directly without standardisation or numeric-only filtering. GReaT linearises tabular rows into token sequences and generates features autoregressively using a pretrained language model; therefore, both continuous and categorical variables were retained in their original coding.

For PredLLM and TapTap, inputs were restricted to numeric variables, as required by their implementations. All numeric columns were standardised using z-score normalisation to stabilise training and to reduce token sequence length with values rounded to three decimal places. For PredLLM, a designated target variable (i.e. the latent SMP score) was positioned as the final column to enable regression conditioning. Synthetic samples were generated in the standardised space and inverse-transformed back to the original scale prior to evaluation.

For REaLTabFormer, numeric variables were similarly standardised to improve convergence and control token length. Although the model supports mixed data types numeric-only inputs were used to maintain comparability across generators. Inverse scaling was applied after sampling.

For Tabula, non-standardised values were retained to preserve ordinal and categorical semantics and categorical variables were explicitly specified. Model selection was guided by a custom SEM-driven callback that evaluated structural path consistency between real and synthetic data during training.

Across all methods, multiple synthetic replicates were generated to assess variability and stability. Any additional post-generation processing applied uniformly across generators is described in a subsequent section.

7.4 Appendix D – Reporting Metrics

This section defines all quantitative metrics used to compare each synthetic-data technique against the real dataset (REAL). Unless stated otherwise, comparisons are performed **pairwise (synthetic vs REAL)** and then **aggregated across replicates** of each technique.

Notation

- $t \in \mathcal{T}$: synthetic technique (REAL excluded from ranking).

- r belongs to $(1, \dots, R_t)$: replicate index for technique t .
- X^{real} : real dataset; $X^{(t,r)}$: synthetic dataset for technique t , replicate r .
- For any scalar metric m , let m^{real} be the value on REAL and $m^{(t,r)}$ the value on a synthetic replicate.
- “Signed delta” is always defined as:

$$\Delta m^{(t,r)} = m^{(t,r)} - m^{\text{real}}.$$

- “Absolute error” is:

$$|\Delta m^{(t,r)}| = |m^{(t,r)} - m^{\text{real}}|.$$

- When a metric is computed per variable/path/pair, we report **means**, **medians**, **maxima**, and/or **ranges** across those items as specified below.

1) Distribution fidelity metrics

Distribution metrics compare marginal statistics and basic validity constraints between REAL and each synthetic technique.

1.1 Continuous distribution deviation (standardized)

For each continuous variable j , with real standard deviation $\sigma_j^{\text{real}} > 0$:

- Standardized mean deviation:

$$\delta_{\mu,j}^{(t)} = \frac{|\mu_j^{(t)} - \mu_j^{\text{real}}|}{\sigma_j^{\text{real}}}$$

- Standardized standard-deviation deviation:

$$\delta_{\sigma,j}^{(t)} = \frac{|\sigma_j^{(t)} - \sigma_j^{\text{real}}|}{\sigma_j^{\text{real}}}$$

These are aggregated across continuous variables:

$$\text{Mean_} |\Delta \mu|^{(t)} = \frac{1}{J_c} \sum_{j=1}^{J_c} \delta_{\mu,j}^{(t)}, \text{Mean_} |\Delta \sigma|^{(t)} = \frac{1}{J_c} \sum_{j=1}^{J_c} \delta_{\sigma,j}^{(t)}.$$

1.2 Boundary adherence (continuous validity)

For each continuous variable j , boundary adherence is computed by comparing whether synthetic values lie within the empirical real-data support:

$$BA_j^{(t)} = \Pr(\min(X_j^{\text{real}}) \leq X_j^{(t)} \leq \max(X_j^{\text{real}})).$$

(Implementation uses SDMetrics BoundaryAdherence.)

Aggregate:

$$\text{Mean_Continuous_Adherence}^{(t)} = \frac{1}{J_c} \sum_{j=1}^{J_c} BA_j^{(t)}.$$

Additionally, the report includes per-variable summary statistics:

$$\mu_j^{(t)}, \sigma_j^{(t)}$$

and difference vs REAL for boundary tables:

$$\Delta\mu_j^{(t)} = \mu_j^{(t)} - \mu_j^{\text{real}}, \Delta\sigma_j^{(t)} = \sigma_j^{(t)} - \sigma_j^{\text{real}}.$$

1.3 Category adherence (categorical/ordinal validity)

For each categorical or ordinal variable k , category adherence measures whether synthetic values remain within the set observed in REAL:

$$CA_k^{(t)} = \Pr(X_k^{(t)} \in \mathcal{C}_k^{\text{real}}), \mathcal{C}_k^{\text{real}} = \{x: x \text{ observed in REAL}\}.$$

(Implementation uses SDMetrics CategoryAdherence.)

Aggregate across categorical/ordinal variables:

$$\text{Mean_Category_Adherence}^{(t)} = \frac{1}{J_o} \sum_{k=1}^{J_o} CA_k^{(t)}.$$

1.4 Range sanity check

For each continuous variable j , after rounding endpoints to 2 decimals:

$$\begin{aligned} \text{Same_As_REAL}_j^{(t)} = \\ \mathbf{1}(\text{round}(\min X_j^{(t)}, 2) = \text{round}(\min X_j^{\text{real}}, 2)) \end{aligned}$$

$$\wedge \text{round}(\max X_j^{(t)}, 2) = \text{round}(\max X_j^{\text{real}}, 2))$$

1.5 Composite distribution score used in ranking

The ranking stage forms a single distribution error (lower is better) for each technique:

$$\begin{aligned} \text{Distribution_Error}^{(t)} = & \text{Mean_} |\Delta\mu|^{(t)} + \text{Mean_} \\ & |\Delta\sigma|^{(t)} + (1 - \text{Mean_Continuous_Adherence}^{(t)}) + (1 \\ & - \text{Mean_Category_Adherence}^{(t)}), \end{aligned}$$

and converts it to a “higher-is-better” score:

$$\text{Distribution_Score}^{(t)} = -\text{Distribution_Error}^{(t)}.$$

All distribution-based deviations are non-negative error quantities with a theoretical minimum of 0 and no finite upper bound. A value of 0 indicates perfect agreement with the real data distribution. Consequently, smaller values indicate better fidelity.

Boundary adherence and category adherence are proportions bounded in the unit interval [0, 1], where 1 indicates that all synthetic values lie within the empirical support. For these metrics, higher values are preferable.

The composite Distribution_Error is therefore bounded below by 0 and increases with worsening fidelity; hence lower values are better. Because ranking is performed on Distribution_Score = -Distribution_Error, larger scores correspond to better performance, with the theoretical optimum at 0.

2) Correlation preservation metrics

2.1 Latent construct correlation preservation (PLS and CB-SEM)

For latent construct pair (a, b) , define the signed correlation delta:

$$\Delta r_{ab}^{(t)} = r_{ab}^{(t)} - r_{ab}^{\text{real}}.$$

This study reports heatmaps of signed Δr .

A stability summary is computed per technique across the selected latent pairs $\mathcal{P}$:

$$\text{Mean } |\Delta r|^{(t)} = \frac{1}{|\mathcal{P}|} \sum_{(a,b) \in \mathcal{P}} |\Delta r_{ab}^{(t)}|,$$

$$\text{Median } |\Delta r|^{(t)} = \text{median}_{(a,b) \in \mathcal{P}} (|\Delta r_{ab}^{(t)}|), \text{Max } |\Delta r|^{(t)} = \max_{(a,b) \in \mathcal{P}} |\Delta r_{ab}^{(t)}|.$$

For ranking, an error is derived from the mean absolute delta (averaged across PLS and CB-SEM):

$$\begin{aligned} \text{LatentCorr_Error}^{(t)} &= \frac{1}{2} \left(\text{Mean } |\Delta r|_{\text{PLS}}^{(t)} + \text{Mean } |\Delta r|_{\text{CB}}^{(t)} \right), \text{LatentCorr_Score}^{(t)} \\ &= -\text{LatentCorr_Error}^{(t)}. \end{aligned}$$

2.2 Covariate–latent correlation preservation

For covariate x and target latent y *belongs to all latent constructs*:

$$\Delta r_{x,y}^{(t)} = r_{x,y}^{(t)} - r_{x,y}^{\text{real}}, |\Delta r_{x,y}^{(t)}| = |r_{x,y}^{(t)} - r_{x,y}^{\text{real}}|.$$

Technique-level summary statistics over all covariate–target pairs $\mathcal{Q}$:

$$\text{MeanAbsDelta}_r^{(t)} = \frac{1}{|\mathcal{Q}|} \sum_{(x,y) \in \mathcal{Q}} |\Delta r_{x,y}^{(t)}|,$$

with analogous median and max reported in the main error-summary table.

Ranking score:

$$\text{Covariate_Score}^{(t)} = -\text{MeanAbsDelta}_r^{(t)}.$$

Correlation preservation is quantified using absolute differences between synthetic and real correlations. Since correlations lie within $[-1, 1]$, the absolute deviation $|\Delta r|$ is bounded between 0 and 2, where 0 denotes perfect preservation. All aggregated correlation deviations are therefore error measures for which smaller values indicate better structural fidelity.

For ranking, these errors are negated to form LatentCorr_Score and Covariate_Score, such that higher scores reflect closer agreement with the real correlation structure.

3) Measurement model fidelity metrics

3.1 Factor loadings (PLS)

For indicator i belonging to construct c , absolute loading error:

$$|\Delta \lambda_i^{(t)}| = |\lambda_i^{(t)} - \lambda_i^{\text{real}}|.$$

Mean absolute loading error by construct:

$$\text{MAE}_c^{(t)} = \frac{1}{|I_c|} \sum_{i \in I_c} |\Delta \lambda_i^{(t)}|.$$

A full **indicator × technique matrix** of $|\Delta \lambda_i^{(t)}|$ is reported, with a heatmap.

Ranking uses a technique-level loading error:

$$\text{Loading_Error}^{(t)} = \frac{1}{C} \sum_{c=1}^C \text{MAE}_c^{(t)}, \text{Loading_Score}^{(t)} = -\text{Loading_Error}^{(t)}.$$

3.2 Reliability and convergent validity

For each construct c , the reliability estimates are:

- Cronbach's alpha: α_c
- Composite reliability: $\rho_{C,c}$
- Dijkstra–Henseler's rho: $\rho_{A,c}$
- Average variance extracted: AVE_c

Threshold compliance (per construct)

Indicators are computed as:

$$\mathbf{1}(\alpha_c \geq 0.70), \mathbf{1}(\rho_{C,c} \geq 0.70), \mathbf{1}(\rho_{A,c} \geq 0.70), \mathbf{1}(\text{AVE}_c \geq 0.50),$$

and displayed using ✓/✗ (REAL is included as baseline).

Mean absolute deviation vs REAL

For metric m belongs to $(\alpha, \rho_C, \rho_A, \text{AVE})$:

$$|\Delta m_c^{(t)}| = |m_c^{(t)} - m_c^{\text{real}}|.$$

These deviations are reported per construct and technique.

Ranking uses:

$$\text{Reliability_Error}^{(t)} = \frac{1}{N} \sum_{c,m} |\Delta m_c^{(t)}|, \text{Reliability_Score}^{(t)} = -\text{Reliability_Error}^{(t)}.$$

Absolute loading deviations $|\Delta\lambda|$ are bounded below by 0 and may theoretically approach 2 when synthetic and real loadings have opposite signs and maximal magnitude. A value of 0 indicates identical factor loadings. Lower values therefore represent superior measurement fidelity.

Reliability and convergent validity coefficients lie within $[0, 1]$. However, evaluation focuses on absolute deviations from the real data estimates. These deviations are minimised at 0 indicating perfect reproduction of construct reliability. The aggregated Reliability_Error is therefore an error metric and its negated score is used for ranking.

4) Discriminant validity metrics

4.1 HTMT maximum and threshold compliance

For each construct pair (c, k) , HTMT is computed (from the HTMT matrix). The report summarizes the maximum:

$$\text{MaxHTMT}^{(t)} = \max_{c < k} \text{HTMT}_{c,k}^{(t)}.$$

Threshold compliance indicator:

$$\mathbf{1}(\text{MaxHTMT}^{(t)} < \tau_{\text{HTMT}}),$$

where τ_{HTMT} is the configured HTMT threshold.

Validity score used in ranking

The ranking converts discriminant validity to a binary score from the $\checkmark/\times$ HTMT criterion:

$$\text{HTMT_Score}^{(t)} \text{ is either } 0 \text{ or } 1.$$

4.2 Error grids and RMSE (Fornell–Larcker and HTMT matrices)

Let M^{real} be the REAL matrix (FL or HTMT) and $M^{(t)}$ the synthetic matrix for technique t on the same construct ordering. Define elementwise error:

$$E^{(t)} = M^{\text{real}} - M^{(t)}, \text{AE}^{(t)} = |E^{(t)}|.$$

RMSE is computed as:

$$RMSE(t) = \sqrt{\text{mean}((E_{ij}^{(t)})^2)},$$

with the diagonal optionally excluded for HTMT (as implemented):

$$E_{ii}^{(t)} \text{ set to NaN before averaging (HTMT).}$$

HTMT values are non-negative and ideally remain below the specified threshold. Lower HTMT values imply stronger discriminant validity. In the ranking stage, discriminant validity is represented as a binary compliance indicator bounded in 0 and 1. 1 denotes threshold satisfaction and therefore better performance.

Matrix reconstruction errors for both Fornell–Larcker and HTMT matrices are summarised using RMSE, which has a theoretical minimum of 0 and no finite upper bound. As a pure error metric, smaller RMSE values indicate closer reproduction of the real discriminant validity structure.

5) Structural model fidelity metrics

5.1 Standardized structural path coefficients (β)

For path p , standardized coefficient:

$$\beta_p^{(t)} \text{ and } \beta_p^{\text{real}}.$$

The report table displays coefficients and significance markers:

$$\beta_p^{(t)} \text{ with } \begin{cases} ** & p\text{-value} < 0.01 \\ * & p\text{-value} < 0.05 \\ \text{ns} & \text{otherwise} \end{cases}$$

Path interval overlap vs REAL (replicate stability)

Using bootstrap SD from the range sheets, synthetic interval:

$$I_p^{(t)} = [\beta_p^{(t)} - s_p^{(t)}, \beta_p^{(t)} + s_p^{(t)}],$$

containment indicator:

$$\mathbf{1}(\beta_p^{\text{real}} \in I_p^{(t)}).$$

5.2 Directional consistency

For each path p , define sign agreement:

$$\mathbf{1}(\text{sign}(\beta_p^{(t)}) = \text{sign}(\beta_p^{\text{real}})).$$

Directional consistency is the proportion of paths with matching sign:

$$\text{DirCons}^{(t)} = \frac{1}{|P|} \sum_{p \in P} \mathbf{1}(\text{sign}(\beta_p^{(t)}) = \text{sign}(\beta_p^{\text{real}})).$$

5.3 Rank preservation (Spearman's ρ on absolute effects)

Let ranks be computed on absolute coefficients:

$$\text{rank}_p^{\text{real}} = \text{rank}(|\beta_p^{\text{real}}|), \text{rank}_p^{(t)} = \text{rank}(|\beta_p^{(t)}|).$$

Rank preservation is Spearman's correlation:

$$\rho^{(t)} = \text{SpearmanCorr}(\text{rank}_p^{\text{real}}, \text{rank}_p^{(t)}).$$

Structural path score used in ranking

$$\text{Path_Score}^{(t)} = \text{DirCons}^{(t)} + \rho^{(t)}.$$

Directional consistency is defined as a proportion of correctly matched coefficient signs and is bounded within [0, 1], with 1 indicating complete agreement. Rank preservation is computed as Spearman's correlation and lies within [-1, 1], where 1 indicates perfect preservation of relative effect magnitudes.

The combined Path_Score, defined as the sum of these two measures, therefore ranges from -1 to 2. Larger values represent superior recovery of the structural relationships among constructs.

6) Indirect (mediation) effects fidelity

For each indirect effect with IV x , mediator m , DV y , the indirect coefficient is $\beta_{x \rightarrow m \rightarrow y}^{\text{ind}}$.

Absolute error:

$$|\Delta \beta^{\text{ind}}|^{(t)} = |\beta^{\text{ind},(t)} - \beta^{\text{ind,real}}|.$$

Significance agreement ("Sig Match") compares the binary significance decision:

$$\text{SigMatch}^{(t)} = \mathbf{1}([p^{\text{real}} < 0.05] = [p^{(t)} < 0.05]).$$

Ranking aggregates indirect-effect error across all indirect paths:

$$\text{Indirect_Error}^{(t)} = \frac{1}{|J|} \sum_{i \in J} |\Delta \beta_i^{\text{ind}}|^{(t)}, \text{Indirect_Score}^{(t)} = -\text{Indirect_Error}^{(t)}.$$

Absolute indirect-effect deviations are non-negative and minimised at 0 with no finite upper bound. Lower values indicate closer agreement with real mediation effects. The ranking procedure negates this error to produce Indirect_Score where higher values correspond to better performance.

Significance agreement is expressed as a binary indicator in (0, 1), where 1 denotes matching inference decisions.

7) CB-SEM global fit and latent R^2

7.1 Global fit indices (mean across replications)

The report provides (per technique) mean fit indices:

- RMSEA, SRMR, CFI, TLI, AIC.

For ranking, a **threshold-count** fit score is computed using four thresholded indices:

$$\text{Fit_Score}^{(t)} = \mathbf{1}(\text{CFI}^{(t)} \geq 0.90) + \mathbf{1}(\text{TLI}^{(t)} \geq 0.90) + \mathbf{1}(\text{RMSEA}^{(t)} \leq 0.08) + \mathbf{1}(\text{SRMR}^{(t)} \leq 0.08).$$

7.2 Latent R^2 (mean)

For endogenous latent z :

$$\mu(R_z^2)^{(t)} = \frac{1}{R_t} \sum_{r=1}^{R_t} R_z^{2,(t,r)}.$$

(Reported for the specified latent constructs in the CB-SEM output.)

Fit indices follow standard SEM conventions. RMSEA and SRMR are non-negative and are minimised, whereas CFI and TLI are bounded within [0, 1] and are maximised. The composite Fit_Score counts the number of indices meeting recommended thresholds and therefore ranges from 0 to 4 with higher values indicating better overall fit.

Latent R^2 values lie within $[0, 1]$ and fidelity is assessed relative to the real baseline rather than by absolute maximisation.

8) Privacy risk metrics

All privacy metrics are computed between REAL and each synthetic replicate and then averaged across replicates per technique.

8.1 Exact Match Rate (EMR)

$$\text{EMR}^{(t)} = \frac{1}{n^{(t)}} \sum_{i=1}^{n^{(t)}} \mathbf{1} \left(x_i^{(t)} \in X^{\text{real}} \right),$$

where exact matching is across all columns (row-wise tuple equality).

8.2 Nearest-neighbour distance

Using numeric columns only let distances from each synthetic row to all real rows be $d(x_i^{(t)}, x_j^{\text{real}})$. The reported statistic is the mean closest distance:

$$\text{NNDR}^{(t)} = \frac{1}{n^{(t)}} \sum_{i=1}^{n^{(t)}} \min_j d(x_i^{(t)}, x_j^{\text{real}}).$$

8.3 Membership inference proxy

Let $m_i^{(t)} = \min_j d(x_i^{(t)}, x_j^{\text{real}})$. Define ε as the 1st percentile of $\{m_i^{(t)}\}$:

$$\varepsilon = \text{Percentile}_1 \left(\{m_i^{(t)}\}_{i=1}^{n^{(t)}} \right).$$

Membership inference proxy:

$$\text{MIA}^{(t)} = \frac{1}{n^{(t)}} \sum_{i=1}^{n^{(t)}} \mathbf{1} \left(m_i^{(t)} < \varepsilon \right).$$

8.4 Distance to closest record (DCR)

Using the same $m_i^{(t)}$:

$$\text{DCR}_{\text{mean}}^{(t)} = \frac{1}{n^{(t)}} \sum_{i=1}^{n^{(t)}} m_i^{(t)}, \text{DCR}_{\text{min}} = \min_i m_i^{(t)}, \text{DCR}_{p05}^{(t)} = \text{Percentile}_5(m_i^{(t)}).$$

8.5 k-anonymity distance proxy

Let $d_{i,(k)}^{(t)}$ be the distance from synthetic row i to its k -th nearest real neighbour:

$$d_{i,(k)}^{(t)} = k\text{-th smallest element of } d(x_i^{(t)}, x_j^{\text{real}}) \}_j.$$

Reported summary:

$$d_{(k)}^{-(t)} = \frac{1}{n^{(t)}} \sum_{i=1}^{n^{(t)}} d_{i,(k)}^{(t)}.$$

Privacy score used in ranking

$$\text{Privacy_Score}^{(t)} = -\text{EMR}^{(t)} - \text{MIA}^{(t)} + \text{NNDR}^{(t)} + \text{DCR}_{p05}^{(t)}.$$

(So lower EMR/MIA and higher NN distance / DCR percentile improve the score.)

Privacy metrics are interpreted in the opposite direction to fidelity metrics. Exact Match Rate (EMR) and Membership Inference risk (MIA) are proportions bounded in $[0, 1]$. Lower values indicate reduced memorisation and therefore stronger privacy protection.

Distance-based measures are non-negative and unbounded above. Larger distances imply greater separation between synthetic and real records and hence improved privacy.

The composite Privacy_Score combines these components such that higher values consistently represent lower disclosure risk.

9) Composite scoring and final ranking

All component “scores” above are converted into a common scale using min–max normalization within the set of synthetic techniques (REAL excluded).

For any technique-level score column S :

$$\text{minmax}(S^{(t)}) = \begin{cases} 0 & \text{max}(S) = \text{min}(S) \\ \frac{S^{(t)} - \text{min}(S)}{\text{max}(S) - \text{min}(S)} & \text{otherwise.} \end{cases}$$

9.1 Overall composite score

The final overall score is a weighted sum of min–max normalized component scores:

$$\text{Overall_Score}^{(t)} = 0.13 * \text{minmax}(\text{Distribution_Score}^{(t)}) + 0.13 * \text{minmax}(\text{Path_Score}^{(t)}) + 0.08 * \text{minmax}(\text{LatentCorr_Score}^{(t)}) + 0.08 * \text{minmax}(\text{Covariate_Score}^{(t)}) + 0.12 * \text{minmax}(\text{LoadingScore}^{(t)}) + 0.10 * \text{minmax}(\text{Reliability_Score}^{(t)}) + 0.12 * \text{minmax}(\text{HTMT_Score}^{(t)}) + 0.09 * \text{minmax}(\text{Fit_Score}^{(t)}) + 0.15 * \text{minmax}(\text{Privacy_Score}^{(t)}) + 0.08 * \text{minmax}(\text{Indirect_Score}^{(t)}).$$

9.2 Overall rank and component-wise ranks

For each component score S , a rank is computed with “higher is better”:

$$\text{Rank}_S^{(t)} = \text{rank}_{\text{desc}}(S^{(t)}),$$

where Rank 1 indicates the best-performing technique (ties use minimum rank).

Overall rank is obtained by sorting techniques by $\text{Overall_Score}^{(t)}$ in descending order:

$$\text{Overall_Rank}^{(t)} = 1 + \text{number of } u \in \mathcal{T}: \text{Overall_Score}^{(u)} > \text{Overall_Score}^{(t)}.$$

All component scores are transformed using min–max normalisation to the interval [0, 1] ensuring comparability across heterogeneous metrics. After transformation, larger values uniformly represent better performance. The weighted sum therefore yields an Overall_Score bounded within [0, 1], where 1 corresponds to the best-performing synthetic technique.

7.5 Appendix E – Supporting Results

7.5.1 PISA

7.5.1.1 Distribution

Table 4

Descriptive Statistics for Continuous Variables (PISA): Real and Synthetic Data

Variable	Metric	REAL	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
AGE	Mean	15.78	15.89	15.77	15.75	15.80	15.78	15.78	15.75	15.75
AGE	SD	0.28	0.29	0.29	0.28	0.29	0.32	0.27	0.28	0.28
AGE	Range	15 to 16	15 to 16	15 to 16	15 to 16	15 to 16	15 to 16	15 to 16	15 to 16	15 to 16
ESCS	Mean	0.48	0.46	0.64	0.49	0.36	0.50	0.57	0.65	0.49
ESCS	SD	0.84	0.95	0.66	0.76	0.91	0.88	0.74	0.65	0.77
ESCS	Range	-2 to 2	-2 to 2	-2 to 2	-2 to 2	-3 to 2	-2 to 2	-2 to 2	-2 to 2	-2 to 1
MCLSIZE	Mean	29.48	29.18	29.60	30.31	27.10	29.38	29.32	29.73	29.97
MCLSIZE	SD	8.01	7.81	8.06	7.95	7.87	8.20	8.02	8.06	7.99
MCLSIZE	Range	18 to 38	18 to 38	18 to 38	13 to 40	14 to 37	18 to 38	18 to 38	18 to 38	18 to 37
PV10MATH	Mean	589.65	580.26	612.38	593.57	565.38	583.41	594.43	609.17	592.07
PV10MATH	SD	100.78	112.09	69.03	95.36	106.21	136.71	92.12	67.32	98.87
PV10MATH	Range	238 to 817	238 to 816	316 to 817	238 to 817	187 to 867	238 to 817	246 to 816	313 to 827	238 to 817
PV1MATH	Mean	588.30	581.14	609.78	591.68	560.09	591.24	587.56	606.21	590.66
PV1MATH	SD	101.41	126.32	69.91	96.79	107.02	134.81	96.13	66.71	98.98

PV1MATH	Range	237 to 943	237 to 916	331 to 816	237 to 943	210 to 908	246 to 942	249 to 936	359 to 838	206 to 816
PV2MATH	Mean	586.83	604.84	609.64	589.80	557.05	582.48	589.56	607.44	589.59
PV2MATH	SD	102.03	116.36	68.16	96.96	109.37	138.46	94.53	66.88	100.93
PV2MATH	Range	277 to 878	277 to 856	277 to 835	286 to 835	260 to 882	277 to 878	277 to 878	329 to 867	209 to 878
PV3MATH	Mean	587.01	593.35	610.73	590.53	562.54	585.36	593.13	606.91	589.86
PV3MATH	SD	100.63	115.02	69.87	96.48	115.04	129.39	91.94	68.23	98.84
PV3MATH	Range	282 to 838	282 to 838	287 to 838	277 to 838	287 to 879	282 to 838	290 to 837	296 to 843	282 to 838
PV4MATH	Mean	585.36	587.55	609.75	589.95	562.14	577.49	588.36	605.93	587.17
PV4MATH	SD	104.01	127.01	70.60	100.36	120.68	144.93	94.44	67.72	101.95
PV4MATH	Range	201 to 833	201 to 833	286 to 819	201 to 833	177 to 861	201 to 833	211 to 833	297 to 861	203 to 833
PV5MATH	Mean	585.74	586.25	608.97	589.36	558.21	586.16	585.74	606.34	588.33
PV5MATH	SD	103.28	127.45	71.61	98.18	104.56	104.17	95.04	69.30	101.70
PV5MATH	Range	206 to 850	206 to 850	316 to 850	205 to 851	188 to 893	206 to 806	211 to 806	288 to 859	206 to 850
PV6MATH	Mean	588.04	601.41	611.64	591.39	560.08	581.92	591.25	607.80	590.33
PV6MATH	SD	101.43	118.19	68.44	95.34	115.79	137.87	92.33	67.45	99.42
PV6MATH	Range	266 to 851	266 to 851	269 to 851	265 to 851	207 to 861	266 to 851	266 to 851	335 to 834	258 to 851
PV7MATH	Mean	589.52	604.11	611.91	593.62	563.38	587.45	594.18	608.23	592.30
PV7MATH	SD	102.24	113.14	71.09	98.12	114.17	140.64	96.59	69.36	101.07
PV7MATH	Range	238 to 876	238 to 841	347 to 876	252 to 875	238 to 867	238 to 876	239 to 874	339 to 833	231 to 881
PV8MATH	Mean	588.60	595.06	612.19	592.46	553.92	584.30	592.49	608.95	592.07
PV8MATH	SD	102.05	114.89	68.41	97.23	116.01	132.48	95.30	68.31	101.08
PV8MATH	Range	293 to 828	293 to 828	296 to 828	265 to 869	285 to 892	293 to 828	295 to 828	238 to 838	211 to 832
PV9MATH	Mean	588.48	594.22	611.10	591.86	554.55	584.33	591.39	608.59	590.67
PV9MATH	SD	101.09	116.63	69.09	95.64	109.64	138.79	92.72	68.04	99.64
PV9MATH	Range	269 to 841	269 to 841	269 to 841	270 to 820	195 to 889	269 to 841	269 to 841	293 to 819	269 to 883
SCHSIZE	Mean	1330.08	1459.01	1347.75	1362.15	1370.00	1332.07	1299.77	1260.82	1268.99
SCHSIZE	SD	832.75	908.73	796.13	718.09	923.69	811.57	787.33	807.30	702.87
SCHSIZE	Range	315 to 4020	315 to 4020	315 to 4020	314 to 4018	206 to 4018	315 to 4020	315 to 4020	3 to 4020	63 to 4019

Note. Mean, standard deviation, and observed range are reported for real and synthetic data.

Table 5

Continuous Distribution Fidelity Summary

Metric	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
Mean $ \Delta\mu $	0.10	0.18	0.04	0.24	0.03	0.03	0.17	0.04
Mean $ \Delta\sigma $	0.14	0.25	0.05	0.09	0.24	0.07	0.26	0.03
Mean Continuous Adherence	1.0000	1.0000	0.9992	0.9978	1.0000	1.0000	0.9998	0.9894

Note. Metrics summarize standardized deviations in mean and standard deviation, and boundary adherence.

Table 6

Category Adherence for Categorical Variables (PISA): Real and Synthetic Data

Variable	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
IMMIG	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
MISCED	1.0000	1.0000	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000
SC064Q01TA	0.3579	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
SC064Q02TA	0.4379	0.9998	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
SC064Q03TA	0.8386	1.0000	0.0000	0.0000	1.0000	1.0000	0.9995	0.0000
SC064Q04NA	0.7636	1.0000	0.0000	0.0000	1.0000	1.0000	0.9998	0.0000
SC064Q05WA	0.4089	0.9998	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
SC064Q06WA	0.2236	1.0000	0.0000	0.0000	1.0000	1.0000	0.9993	0.0000
SC064Q07WA	0.7196	1.0000	0.0000	0.0000	1.0000	1.0000	0.9988	0.0000
ST001D01T	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ST004D01T	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ST268Q01JA	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ST268Q04JA	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ST268Q07JA	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Note. Values report category adherence for synthetic techniques only.

7.5.1.2 Covariate Correlations

Table 7

Covariate Correlation Error Summary Across Techniques

Metric	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
Mean Δr	0.031	0.044	0.043	0.057	0.016	0.025	0.034	0.033
Median Δr	0.027	0.037	0.031	0.053	0.013	0.023	0.025	0.020
Max Δr	0.098	0.162	0.174	0.135	0.055	0.080	0.110	0.143

Note. Values report mean, median, and maximum absolute correlation deviations aggregated across all latent constructs.

7.5.1.3 Reliability

Table 8

Mean Absolute Deviation of Reliability Metrics from Real Data

Metric	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFmr
SPI – Absolute error (α)	0.07	0.04	0.09	0.01	0.05	0.04	0.04	0.03
SPI – Absolute error (ρC)	0.04	0.10	0.05	0.08	0.08	0.02	0.04	0.22
SPI – Absolute error (ρA)	0.19	0.04	0.06	0.04	0.15	0.06	0.02	0.09
SPI – Absolute error (AVE)	0.06	0.01	0.01	0.01	0.03	0.01	0.00	0.04
SMS – Absolute error (α)	0.01	0.04	0.06	0.00	0.02	0.09	0.04	0.10
SMS – Absolute error (ρC)	0.01	0.02	0.02	0.00	0.01	0.04	0.02	0.05
SMS – Absolute error (ρA)	0.09	0.00	0.07	0.04	0.03	0.05	0.02	0.04
SMS – Absolute error (AVE)	0.02	0.03	0.02	0.00	0.01	0.06	0.03	0.06
SMP – Absolute error (α)	0.02	0.01	0.00	0.00	0.00	0.00	0.10	0.00
SMP – Absolute error (ρC)	0.02	0.01	0.00	0.00	0.00	0.00	0.08	0.00
SMP – Absolute error (ρA)	0.02	0.01	0.00	0.00	0.00	0.00	0.09	0.00
SMP – Absolute error (AVE)	0.16	0.05	0.02	0.02	0.01	0.01	0.41	0.03

Note. Absolute Error values represent absolute deviations between synthetic and real reliability estimates.

7.5.1.4 Discriminant Validity

Table 9

RMSE of Fornell–Larcker and HTMT Error Grids

Metric	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
RMSE (Fornell–Larcker)	0.0547	0.0231	0.0436	0.0307	0.0335	0.0324	0.1000	0.0624
RMSE (HTMT)	0.0691	0.0242	0.0595	0.0458	0.0446	0.0403	0.0280	0.0803

Note. RMSE values summarize absolute error between synthetic and real discriminant validity matrices.

7.5.1.5 Structural Paths

Table 10

Standardized Structural Path Coefficients

Path	REAL β	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
SPI -> SMS	-0.053 ns	-0.038 ±	-0.107 ±	-0.012 ±	-0.015 ±	-0.098 ±	-0.135 ±	-0.088 ±	-0.276 ± 0.060 **
		0.070 ns	0.080 ns	0.090 ns	0.070 ns	0.070 ns	0.040 *	0.040 ns	
SPI -> SMP	-0.207 **	-0.374 ±	-0.211 ±	-0.119 ±	-0.116 ±	-0.232 ±	-0.205 ±	-0.254 ±	-0.425 ± 0.100 **
		0.050 **	0.100 *	0.080 ns	0.040 **	0.050 **	0.030 **	0.030 **	
gender -> SMS	0.121 **	0.058 ±	0.068 ±	0.075 ±	0.092 ±	0.071 ±	0.108 ±	0.137 ±	0.191 ± 0.050 **
		0.040 ns	0.040 ns	0.040 ns	0.030 *	0.040 ns	0.040 *	0.040 **	

gender -> SMP	-0.042 ns	-0.058 ± 0.030 ns	0.035 ± 0.030 ns	0.003 ± 0.030 ns	-0.027 ± 0.030 ns	-0.054 ± 0.030 ns	-0.023 ± 0.030 ns	-0.025 ± 0.030 ns	-0.132 ± 0.030 **
grade -> SMS	0.087 *	0.101 ± 0.040 ns	0.072 ± 0.040 ns	0.032 ± 0.040 ns	0.071 ± 0.040 ns	0.059 ± 0.040 ns	0.039 ± 0.040 ns	0.097 ± 0.040 *	0.024 ± 0.040 ns
grade -> SMP	0.122 **	0.045 ± 0.030 ns	0.141 ± 0.040 **	0.091 ± 0.040 *	0.119 ± 0.030 **	0.074 ± 0.030 ns	0.125 ± 0.040 **	0.145 ± 0.030 **	0.094 ± 0.030 *
motherEdu -> SMS	0.032 ns	0.014 ± 0.040 ns	0.008 ± 0.050 ns	0.053 ± 0.040 ns	-0.036 ± 0.050 ns	0.041 ± 0.050 ns	0.044 ± 0.050 ns	0.029 ± 0.040 ns	0.065 ± 0.060 ns
motherEdu -> SMP	-0.145 **	0.007 ± 0.040 ns	-0.026 ± 0.050 ns	0.083 ± 0.040 *	-0.095 ± 0.050 ns	-0.049 ± 0.050 ns	-0.077 ± 0.040 ns	-0.014 ± 0.040 ns	0.008 ± 0.050 ns
ESCS -> SMS	0.049 ns	-0.072 ± 0.040 ns	0.005 ± 0.050 ns	0.033 ± 0.040 ns	0.241 ± 0.050 **	0.032 ± 0.050 ns	0.073 ± 0.050 ns	0.045 ± 0.040 ns	-0.061 ± 0.070 ns
ESCS -> SMP	0.380 **	0.241 ± 0.030 **	0.209 ± 0.050 **	0.219 ± 0.040 **	0.434 ± 0.040 **	0.284 ± 0.050 **	0.341 ± 0.040 **	0.262 ± 0.040 **	0.242 ± 0.050 **
age -> SMS	-0.027 ns	0.092 ± 0.040 ns	-0.017 ± 0.040 ns	0.012 ± 0.040 ns	0.013 ± 0.040 ns	0.003 ± 0.040 ns	-0.016 ± 0.040 ns	-0.066 ± 0.040 ns	-0.059 ± 0.050 ns
age -> SMP	0.027 ns	0.054 ± 0.030 ns	0.025 ± 0.030 ns	0.016 ± 0.030 ns	0.040 ± 0.030 ns	-0.018 ± 0.030 ns	0.023 ± 0.030 ns	0.019 ± 0.030 ns	-0.018 ± 0.030 ns
immig -> SMS	0.091 *	0.075 ± 0.040 ns	0.028 ± 0.050 ns	0.048 ± 0.040 ns	0.036 ± 0.040 ns	0.071 ± 0.040 ns	0.147 ± 0.050 *	0.146 ± 0.050 **	0.079 ± 0.040 ns
immig -> SMP	0.035 ns	0.086 ± 0.030 **	-0.033 ± 0.040 ns	-0.060 ± 0.040 ns	0.032 ± 0.030 ns	0.096 ± 0.030 ns	0.031 ± 0.040 ns	0.041 ± 0.040 ns	0.102 ± 0.030 **
classSize -> SMS	0.014 ns	0.092 ± 0.040 *	-0.038 ± 0.040 ns	0.006 ± 0.040 ns	0.027 ± 0.040 ns	0.051 ± 0.040 ns	0.000 ± 0.040 ns	0.043 ± 0.040 ns	0.121 ± 0.040 ns
classSize -> SMP	0.092 **	0.048 ± 0.030 *	0.087 ± 0.030 *	0.079 ± 0.040 *	0.080 ± 0.030 *	0.057 ± 0.030 ns	0.100 ± 0.030 **	0.091 ± 0.030 ns	0.151 ± 0.040 **
schoolSize -> SMS	-0.112 **	-0.006 ± 0.040 ns	-0.072 ± 0.040 ns	-0.025 ± 0.040 ns	-0.008 ± 0.040 ns	0.036 ± 0.040 ns	-0.164 ± 0.050 **	-0.138 ± 0.050 **	-0.126 ± 0.050 *
schoolSize -> SMP	0.059 ns	0.011 ± 0.030 ns	0.029 ± 0.040 ns	0.072 ± 0.040 ns	-0.005 ± 0.040 ns	0.011 ± 0.040 ns	0.062 ± 0.040 ns	0.083 ± 0.040 ns	-0.141 ± 0.040 *
SMS -> SMP	0.316 **	0.438 ± 0.030 **	0.283 ± 0.030 **	0.301 ± 0.030 **	0.319 ± 0.030 **	0.330 ± 0.030 **	0.284 ± 0.030 **	0.312 ± 0.030 **	0.176 ± 0.030 **

Note. Entries report standardized β coefficients; asterisks denote statistical significance (* $p < .05$, ** $p < .01$).

7.5.1.6 Loadings

Table 11

Absolute Loading Differences by Indicator and Technique

Indicator	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
PV10MATH	0.088	0.024	0.011	0.010	0.008	0.007	0.284	0.010
PV1MATH	0.081	0.022	0.006	0.012	0.007	0.003	0.209	0.008
PV2MATH	0.090	0.030	0.011	0.011	0.006	0.003	0.287	0.018
PV3MATH	0.078	0.028	0.007	0.012	0.005	0.001	0.255	0.018
PV4MATH	0.080	0.036	0.013	0.012	0.006	0.008	0.273	0.034
PV5MATH	0.075	0.029	0.011	0.011	0.007	0.006	0.025	0.022
PV6MATH	0.099	0.028	0.011	0.010	0.007	0.003	0.306	0.012
PV7MATH	0.091	0.028	0.004	0.013	0.006	0.007	0.298	0.007
PV8MATH	0.085	0.028	0.010	0.012	0.009	0.004	0.272	0.008
PV9MATH	0.078	0.025	0.006	0.012	0.004	0.002	0.267	0.017
ST268Q01JA	0.086	0.012	0.010	0.014	0.013	0.054	0.016	0.020
ST268Q04JA	0.070	0.005	0.006	0.017	0.015	0.002	0.005	0.017
ST268Q07JA	0.153	0.075	0.080	0.015	0.035	0.073	0.041	0.196
SC064Q01TA	0.041	0.088	0.079	0.147	0.054	0.068	0.088	0.187
SC064Q02TA	0.092	0.041	0.122	0.204	0.258	0.019	0.073	0.093

SC064Q03TA	0.100	0.002	0.168	0.061	0.097	0.029	0.041	0.144
SC064Q04NA	0.019	0.039	0.046	0.154	0.190	0.057	0.084	0.439
SC064Q05WA	0.015	0.134	0.021	0.171	0.169	0.073	0.129	0.290
SC064Q06WA	0.060	0.009	0.013	0.000	0.206	0.013	0.011	0.144
SC064Q07WA	0.122	0.053	0.040	0.048	0.373	0.006	0.024	0.312

Note. Each cell reports the absolute difference between synthetic and real factor loadings.

Table 12

Mean Absolute Loading Error by Construct

Construct	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
SMP	0.085	0.028	0.009	0.011	0.006	0.004	0.248	0.016
SMS	0.103	0.031	0.032	0.015	0.021	0.043	0.021	0.078
SPI	0.064	0.052	0.070	0.112	0.192	0.038	0.064	0.230

Note. Values report the mean absolute difference between synthetic and real factor loadings for each construct.

7.5.1.7 Global Fit Measures

Table 13

CB-SEM Global Fit Indices (Mean Across Replications)

Metric	CTGAN	GReaT	PredLLM	REAL	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
RMSEA (≤ 0.08)	0.07	0.10	0.08	0.09	0.11	0.09	0.10	0.09	0.08
SRMR (≤ 0.08)	0.08	0.09	0.07	0.09	0.10	0.08	0.09	0.09	0.08
CFI (≥ 0.90)	0.88	0.85	0.91	0.89	0.84	0.72	0.88	0.89	0.93
TLI (≥ 0.90)	0.87	0.83	0.90	0.88	0.82	0.68	0.87	0.87	0.92
AIC (↓)	182527.77	165141.11	161222.44	170374.59	168266.25	188667.81	166715.85	154883.38	165116.25

Note. Values report mean global fit indices for each technique. Recommended thresholds are shown in parentheses.

Table 14

Latent Variable R^2 for CB-SEM Structural Paths

Metric	CTGAN	GReaT	PredLLM	REAL	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
Mean R^2 (SMP)	0.393	0.193	0.249	0.289	0.277	0.247	0.310	0.189	0.356
Mean R^2 (SMS)	0.062	0.029	0.041	0.044	0.077	0.066	0.062	0.021	0.060

Note. Values report mean explained variance (R^2) for each latent endogenous construct.

7.5.1.8 Indirect (Mediation) Effects Relative to Real Data

Table 15

Indirect (Mediation) Effects Relative to Real Data

Metric	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
$ \Delta\beta $	0.006	0.013	0.013	0.011	0.010	0.022	0.013	0.030
Sig Match	False	True	True	True	True	False	True	False

Note. Absolute error computed as $|\text{synthetic} - \text{real}|$. Significance agreement indicates whether both datasets agree on $p < .05$.

7.5.2 TIMSS

7.5.2.1 Distribution

Table 16

Descriptive Statistics for Continuous Variables (TIMSS): Real and Synthetic Data

Variable	Metric	REAL	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
----------	--------	------	-------	-------	---------	---------------	---------	--------	--------	--------

BTBG10	Mean	23.39	19.99	25.22	22.02	24.28	23.52	21.99	23.78	22.63
BTBG10	SD	13.16	13.21	13.29	13.17	12.42	13.91	12.93	13.44	12.93
BTBG10	Range	1 to 44	1 to 44	1 to 44	1 to 44	1 to 44	1 to 43	1 to 44	1 to 44	1 to 44
BTBGSOS	Mean	10.97	10.45	11.14	11.10	9.78	11.08	11.10	5.09	11.10
BTBGSOS	SD	2.24	2.13	2.24	2.09	2.12	2.21	2.03	3.52	2.75
BTBGSOS	Range	4 to 13	4 to 13	4 to 13	5 to 13	4 to 13	4 to 13	4 to 13	1 to 18	-59 to 13

Note. Mean, standard deviation, and observed range are reported for real and synthetic data.

Table 17

Continuous Distribution Fidelity Summary

Metric	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
Mean $ \Delta\mu $	0.25	0.11	0.08	0.30	0.03	0.08	1.33	0.06
Mean $ \Delta\sigma $	0.03	0.01	0.03	0.05	0.03	0.05	0.30	0.12
Mean Continuous Adherence	1.0000	1.0000	0.8522	0.9235	1.0000	1.0000	0.7582	0.9938

Note. Metrics summarize standardized deviations in mean and standard deviation, and boundary adherence.

Table 18

Category Adherence for Categorical Variables (TIMSS): Real and Synthetic Data

Variable	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
BTBG06I	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG06J	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG06K	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13A	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13B	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13C	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13D	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13E	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13F	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13G	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13H	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBG13I	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBS17A	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBS17B	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBS17C	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBS17D	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000
BTBGSOS	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000

Note. Values report category adherence for synthetic techniques only.

7.5.2.2 Covariate Correlations

Table 19

Covariate Correlation Error Summary Across Techniques

Metric	CTGAN	GReaT	PredLLM	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
Mean $ \Delta r $	0.062	0.052	0.040	0.044	0.019	0.021	0.161	0.072
Median $ \Delta r $	0.043	0.053	0.020	0.025	0.014	0.014	0.106	0.049
Max $ \Delta r $	0.142	0.089	0.112	0.143	0.042	0.050	0.508	0.165

Note. Values report mean, median, and maximum absolute correlation deviations aggregated across all latent constructs.

7.5.2.3 Reliability

Table 20

Mean Absolute Deviation of Reliability Metrics from Real Data

Construct	Metric	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
SSF	Absolute error (α)	15.67	2.51	16.78	5.10	6.43	5.74	7.70	1.91
SSF	Absolute error (ρ_C)	0.00	0.00	0.13	0.01	0.00	0.00	0.00	0.00
SSF	Absolute error (ρ_A)	0.27	0.01	0.19	0.03	0.02	0.02	0.04	0.01
SSF	Absolute error (AVE)	0.19	0.01	0.31	0.05	0.02	0.02	0.03	0.01
TCI	Absolute error (α)	0.06	0.03	0.00	0.02	0.02	0.00	0.05	0.06
TCI	Absolute error (ρ_C)	0.03	0.02	0.02	0.02	0.01	0.01	0.03	0.10
TCI	Absolute error (ρ_A)	0.12	0.08	0.26	0.06	0.26	0.06	0.10	0.25
TCI	Absolute error (AVE)	0.08	0.07	0.03	0.06	0.03	0.01	0.07	0.19
LEI	Absolute error (α)	0.02	0.01	0.03	0.01	0.02	0.01	0.01	0.01
LEI	Absolute error (ρ_C)	0.02	0.01	0.02	0.00	0.01	0.01	0.01	0.02
LEI	Absolute error (ρ_A)	0.02	0.00	0.02	0.01	0.01	0.01	0.01	0.00
LEI	Absolute error (AVE)	0.04	0.01	0.03	0.00	0.01	0.01	0.01	0.02
ACM	Absolute error (α)	0.12	0.03	0.08	0.00	0.01	0.01	0.02	0.05
ACM	Absolute error (ρ_C)	0.06	0.01	0.04	0.00	0.00	0.01	0.01	0.03
ACM	Absolute error (ρ_A)	0.12	0.01	0.07	0.00	0.00	0.01	0.02	0.07
ACM	Absolute error (AVE)	0.11	0.03	0.08	0.00	0.01	0.01	0.02	0.06

Note. Absolute Error values represent absolute deviations between synthetic and real reliability estimates.

7.5.2.4 Discriminant Validity

Table 21

RMSE of Fornell–Larcker and HTMT Error Grids

Metric	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
RMSE (Fornell–Larcker)	0.2756	0.0448	0.3925	0.0975	0.0459	0.0266	0.0853	0.2796
RMSE (HTMT)	0.1720	0.0695	0.0696	0.1412	0.0691	0.0395	0.0345	0.0911

Note. RMSE values summarize absolute error between synthetic and real discriminant validity matrices.

7.5.2.5 Structural Paths

Table 22

Standardized Structural Path Coefficients

Path	REAL β	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
SSF -> ACM	-0.478 **	-0.018 $\pm$ 0.450 ns	-0.445 $\pm$ 0.280 *	0.494 $\pm$ 0.070 **	-0.221 $\pm$ 0.270 ns	-0.350 $\pm$ 0.130 **	-0.406 $\pm$ 0.340 *	-0.308 $\pm$ 0.440 ns	0.090 $\pm$ 0.400 ns
TCI -> ACM	0.107 *	0.241 $\pm$ 0.050 **	0.106 $\pm$ 0.040 *	0.116 $\pm$ 0.090 ns	0.107 $\pm$ 0.040 *	0.022 $\pm$ 0.090 ns	0.088 $\pm$ 0.060 ns	0.101 $\pm$ 0.050 *	0.150 $\pm$ 0.060 **
LEI -> ACM	0.304 **	0.338 $\pm$ 0.050 **	0.416 $\pm$ 0.050 **	0.141 $\pm$ 0.050 *	0.484 $\pm$ 0.050 **	0.466 $\pm$ 0.050 **	0.374 $\pm$ 0.040 **	0.293 $\pm$ 0.040 **	0.465 $\pm$ 0.060 **
NSC -> ACM	-0.062 ns	-0.050 $\pm$ 0.040 ns	-0.106 $\pm$ 0.040 *	-0.069 $\pm$ 0.050 ns	-0.096 $\pm$ 0.040 ns	-0.081 $\pm$ 0.050 ns	-0.070 $\pm$ 0.050 ns	-0.040 $\pm$ 0.040 ns	-0.029 $\pm$ 0.040 ns

Note. Entries report standardized β coefficients; asterisks denote statistical significance (* $p < .05$, ** $p < .01$).

7.5.2.6 Loadings

Table 23

Absolute Loading Differences by Indicator and Technique

Indicator	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
BTBG06I	0.083	0.007	0.065	0.007	0.002	0.003	0.027	0.039
BTBG06J	0.081	0.002	0.020	0.004	0.007	0.018	0.010	0.040
BTBG06K	0.033	0.050	0.054	0.011	0.023	0.002	0.003	0.025
BTBG13A	0.005	0.072	0.092	0.007	0.004	0.058	0.006	0.083
BTBG13B	0.022	0.151	0.007	0.105	0.097	0.045	0.077	0.157
BTBG13C	0.051	0.046	0.083	0.075	0.023	0.030	0.030	0.055
BTBG13D	0.084	0.022	0.023	0.046	0.012	0.042	0.049	0.057
BTBG13E	0.043	0.002	0.012	0.017	0.011	0.048	0.005	0.043
BTBG13F	0.004	0.013	0.012	0.004	0.020	0.025	0.019	0.026
BTBG13G	0.008	0.081	0.046	0.039	0.051	0.005	0.001	0.081
BTBG13H	0.040	0.019	0.059	0.024	0.012	0.075	0.021	0.140
BTBG13I	0.059	0.042	0.069	0.032	0.003	0.007	0.070	0.025
BTBG10	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BTBGSOS	1.141	0.002	1.593	0.349	0.010	0.010	0.400	1.171
BDGSOS	1.149	0.003	1.917	0.386	0.016	0.012	0.402	1.169
BTBS17A	0.011	0.035	0.102	0.025	0.026	0.027	0.034	0.055
BTBS17B	0.038	0.080	0.042	0.072	0.043	0.014	0.050	0.220
BTBS17C	0.055	0.065	0.018	0.060	0.008	0.011	0.040	0.233
BTBS17D	0.089	0.044	0.018	0.032	0.010	0.034	0.042	0.150

Note. Each cell reports the absolute difference between synthetic and real factor loadings.

Table 24

Mean Absolute Loading Error by Construct

Construct	CTGAN	GReaT	Tabula	TapTap	PredLLM	TabSyn	TabDiff	REaLTabFormer
ACM	0.066	0.020	0.046	0.007	0.011	0.008	0.013	0.035
TCI	0.048	0.056	0.045	0.047	0.022	0.022	0.042	0.165
LEI	0.035	0.050	0.045	0.039	0.026	0.037	0.031	0.074
NSC	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SSF	1.145	0.003	1.755	0.368	0.013	0.011	0.401	1.170

Note. Values report the mean absolute difference between synthetic and real factor loadings for each construct.

7.5.2.7 Global Fit Measures

Table 25

CB-SEM Global Fit Indices (Mean Across Replications)

Metric	CTGAN	GReaT	PredLLM	REAL	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
RMSEA (≤ 0.08)	0.060	0.070	0.050	0.070	0.110	0.040	0.070	0.060	0.070
SRMR (≤ 0.08)	0.060	0.060	0.050	0.060	0.090	0.050	0.060	0.070	0.060
CFI (≥ 0.90)	0.920	0.930	0.960	0.930	0.820	0.960	0.910	0.920	0.930
TLI (≥ 0.90)	0.910	0.910	0.950	0.920	0.790	0.950	0.890	0.910	0.920
AIC (↓)	14345.04	10505.74	10871.58	12672.89	12230.40	13115.10	12423.24	11379.71	10838.23

Note. Values report mean global fit indices for each technique. Recommended thresholds are shown in parentheses.

Table 26

Latent Variable R^2 for CB-SEM Structural Paths

Metric	CTGAN	GReaT	PredLLM	REAL	REaLTabFormer	TabDiff	TabSyn	Tabula	TapTap
Mean R^2 (ACM)	0.781	0.573	0.476	0.439	0.547	0.501	0.447	0.197	0.568

Note. Values report mean explained variance (R^2) for each latent endogenous construct.

7.6 Appendix F: Rationale for the Use of PLS-SEM as the Primary Method with CB-SEM as a Secondary Diagnostic

This study adopts a hybrid SEM-based evaluation strategy in which PLS-SEM is used as the primary analytical method, with CB-SEM global fit indices reported as a secondary diagnostic, based on the nature of the data and the objectives of the benchmark.

The comparison between real and synthetic data is framed as a mimicry-based evaluation rather than a comparison of independent populations. Synthetic datasets are generated to approximate the structure of the original data, and the evaluation therefore focuses on whether key relational and measurement properties are recovered. In this context, formal multi-group measurement invariance testing is not required for the primary benchmarking objective.

The behavioural and human-centred datasets examined in this study (PISA and TIMSS) comprise largely ordinal, categorical, and constrained indicators specified within theory-driven SEMs. PLS-SEM supports estimation and comparison of indicator–construct relationships, reliability, discriminant validity, and structural path coefficients under these data conditions, enabling consistent comparison across multiple synthetic data generators.

The aim of the study is benchmarking synthetic data generators, not psychometric scale validation. Accordingly, emphasis is placed on downstream analytical reproducibility—that is, whether analyses conducted on synthetic data yield parameter estimates and substantive conclusions consistent with those obtained from real data. This evaluation logic is used in synthetic data research in which fidelity and utility are assessed through downstream task performance (Wang et al., 2025) (Hudovernik et al., 2024).

CB-SEM global fit indices are reported to provide a high-level diagnostic of covariance-level plausibility which complements the primary PLS-SEM analysis. It also helps to identify gross structural distortions whilst remaining within the scope of a benchmarking study rather than a full confirmatory factor analysis.

7.7 Appendix G: Linear SEM Formulation and Implications for Target Construction

Structural Equation Modelling (SEM) provides a framework for representing relationships among latent constructs using systems of linear equations. In the standard linear SEM formulation - the model consists of a structural component

relating latent variables to each other and a measurement component linking latent variables to observed indicators.

Structural Model

Let $\boldsymbol{\eta}$ denote the vector of endogenous latent variables and $\boldsymbol{\xi}$ the vector of exogenous latent variables. The structural relationships are expressed as

$$\boldsymbol{\eta} = B\boldsymbol{\eta} + \Gamma\boldsymbol{\xi} + \boldsymbol{\zeta},$$

where

- B is the matrix of coefficients among endogenous latent variables
- Γ is the matrix of effects from exogenous to endogenous latent variables
- $\boldsymbol{\zeta}$ represents structural disturbance terms

Rearranging,

$$(I - B)\boldsymbol{\eta} = \Gamma\boldsymbol{\xi} + \boldsymbol{\zeta}.$$

Assuming $I - B$ is nonsingular (a standard identification and stability condition), we obtain

$$\boldsymbol{\eta} = (I - B)^{-1}\Gamma\boldsymbol{\xi} + (I - B)^{-1}\boldsymbol{\zeta}.$$

Thus, the endogenous latent variables are linear functions of exogenous latent variables and disturbances.

Measurement Model

Observed indicators are modelled as linear functions of latent variables.

For indicators of endogenous constructs:

$$\mathbf{y} = \Lambda_y\boldsymbol{\eta} + \boldsymbol{\varepsilon},$$

and for indicators of exogenous constructs:

$$\mathbf{x} = \Lambda_x\boldsymbol{\xi} + \boldsymbol{\delta},$$

where

- Λ_y, Λ_x are loading matrices
- $\boldsymbol{\varepsilon}, \boldsymbol{\delta}$ are measurement error terms

Substituting the structural solution for η into the measurement equation yields

$$\mathbf{y} = \Lambda_y(I - B)^{-1}\Gamma\xi + \Lambda_y(I - B)^{-1}\zeta + \epsilon.$$

This expression demonstrates that observed variables are linear functions of exogenous latent variables and disturbance terms.

Implications

Under these assumptions, linear SEM corresponds to a system of linear equations relating latent constructs and observed indicators. Endogenous latent variables play a role analogous to dependent variables in regression models and exogenous constructs function as predictors.

This property provides a theoretical justification for using the composite score of an endogenous latent construct as a target variable. Although the latent construct itself is unobserved - its estimated composite score serves as an empirical proxy that preserves the directionality of the structural relationships.

To maintain consistency with SEM practice - any such target column used during model training is excluded from subsequent SEM analyses of synthetic data and latent constructs are re-estimated from observed indicators.

Scope Note

The derivation above assumes a linear SEM framework without nonlinear effects, interactions, or latent growth components. Extensions of SEM can incorporate nonlinear relationships, but these are not considered in the present study.

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